



Large Language Model Guided Graph Capsule Network with Local-Enhanced Alignment for Fake News Detection

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ABSTRACT

Fake news on social media increasingly employs advanced deceptive strategies such as partial tampering and semantic misleading, which create a camouflaged consistency as it exhibits superficial coherence while harboring deeper inconsistencies. Existing detection methods treat cross-modal consistency as a monolithic concept and apply uniform strategies across all samples, rendering them insensitive to these nuanced deception patterns. To address the above issues, we propose the Large Language Model Guided Graph Capsule Network with Local-Enhanced Alignment (LGCNLA) model, a framework that systematically targets three critical dimensions of deception. First, at the semantic layer, we leverage Large Language Models to generate multi-perspective image-enhanced descriptions and supplement external entities, making implicit misleading cues explicit and providing richer contextual evidence. Second, at the structural layer, we construct a dual-channel heterogeneous graph capsule network whose dynamic routing mechanism preserves and amplifies local anomalous patterns that traditional graph based methods would smooth away. Third, at the distributional layer, we reformulate fake news detection as a hierarchical domain alignment task and introduce Local-Enhanced Maximum Mean Discrepancy, which separately aligns cross-modal distributions for real and fake news by minimizing divergence for authentic content while maximizing it for deceptive content, with locality-aware sensitivity to fine-grained inconsistencies. These three components function synergistically such that semantic enrichment provides the evidence, structural preservation highlights the anomalies, and distributional calibration defines the decision objective, thereby creating a holistic detection system that penetrates the camouflage of advanced deceptive strategies. Extensive experiments on Weibo, PolitiFact, and GossipCop datasets demonstrate that our method outperforms the state-of-the-art methods, with accuracy improvements of at least 0.021, 0.029, and 0.036 respectively. The main code is available at <https://github.com/sygygywaityou/LGCNLA>.

1. Introduction

As an important branch of natural language processing, Fake News Detection (FND) utilizes technologies such as text classification, computer vision and machine learning to automatically identify and determine the authenticity of news content (Shang et al., 2025). With the popularization of social media and digital news platforms, the spread speed and influence of fake news

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Fig. 1. Examples of fake news with advanced deceptive strategies.

have expanded sharply, which not only misleads public perception but also triggers social unrest, affect political elections and even endanger public health. As the volume of news on social platforms such as Facebook, Twitter, and TikTok grows explosively, the importance of automated techniques for FND has become increasingly prominent, which provides continuous motivation and social demand for researchers to develop more intelligent and efficient detection systems (Dai et al., 2025; Wang et al., 2025).

The FND methods mainly include text feature based methods, social context based methods, and multimodal fusion based methods. Text feature based methods (Rubin et al., 2016; Wang, 2021) combine language style, emotional tendencies, semantic contradictions, and other clues to classify through machine learning or deep learning models. Social context based methods (Monti et al., 2019; Shu et al., 2020) focus on analyzing network features such as propagation paths and user interaction patterns. Multimodal fusion based methods (Wang et al., 2023; Xu et al., 2024) combine text, images, videos and other content to capture image text inconsistency through cross modal alignment, attention mechanism or Graph Neural Network (GNN) models.

However, text feature based methods are difficult to identify semantically hidden fake content and have limited generalization ability for cross linguistic or professional news. Social context based methods rely on the integrity of the dissemination data, and their effectiveness significantly decrease in the early stages of news dissemination or in closed social circles. Multimodal fusion based methods lack sensitivity to complex contradictions between modalities such as local tampering and misleading titles, and are difficult to solve the problem of modality inconsistency. To solve the above issues, some researchers (Bazmi et al., 2024; Li et al., 2025) have introduced external knowledge such as knowledge graphs and fact checking libraries to assist in verifying the authenticity of news, while recent works (Chen et al., 2025; Park et al., 2025) have also begun to explore the adversarial application of Large Language Models (LLMs) in the generation and detection of fake news. Hu et al. (2025) propose a multimodal FND method that integrates LLMs with global label propagation. By constructing a cross-modal graph structure based on semantic features extracted by LLMs and iteratively optimizing it through label confidence during propagation, the model's ability to model complex multimodal relationships is effectively enhanced. Zhang et al. (2025) innovatively leverage the hallucination property of LLMs for negative reasoning, generating hypothetical scenarios that contradict the logic of news content to construct contradictory feature vectors. Tong et al. (2025) use LLMs to generate diverse news variants as augmented samples and employ reinforcement learning strategies to adaptively select high-quality samples to expand the training data, significantly improving the model's detection performance and generalization ability in data-scarce scenarios. However, despite the fact that many researchers have carried out extensive FND works, the following problem still exist:

- **Insensitive to advanced deceptive strategies such as partial tampering and semantic misguidance.** Partial tampering refers to the localized modification of news content to create fake information with a shell of authenticity (as shown in Fig. 1(a) and (b)). Semantic misleading is a deceptive strategy that manipulates contextual associations, implied meanings, or emotional tendencies to mislead readers into drawing incorrect conclusions while maintaining surface authenticity (as shown in Fig. 1(c)). Existing LLM based methods typically use pre-trained models to extract features and combine noise filtering techniques such as attention mechanisms to suppress unreliable signals. However, these methods have obvious limitations. Firstly, pre-trained models may perform poorly in the FND tasks as they are typically trained on a general corpus. Secondly, simple attention mechanisms are difficult to distinguish between “deliberately forged signals” and “normal semantic changes”, resulting in the model being insensitive to advanced deception strategies.

To solve the above issues, we propose the Large Language Model Guided Graph Capsule Network with Local-Enhanced Alignment (LGCNLA) model, which tackles the FND problem by systematically decomposing and reinforcing the notion of “cross-modal consistency” from three orthogonal perspectives. Our contributions are not merely an assembly of techniques, but a targeted response to the multi-faceted nature of advanced deception.

Semantic Enrichment (Addressing the “What is implied?” gap). Existing methods often fail to capture deep semantic deceptions because they operate on surface-level features. We argue that the missing piece is contextual intelligence. Therefore, we leverage LLMs not just as feature extractors, but as simulated readers to generate multi-perspective image descriptions, and

as domain experts to supplement latent external entities. This constructs a semantically augmented corpus that makes implicit misleading cues explicit, providing a richer foundation for detecting semantic misguidance.

Anomaly-Preserving Representation (Addressing the “Where is the anomaly?” gap). Standard GNNs, commonly used in multimodal fusion, are inherently smoothing operators that dilute local anomalies like partial tampering. We identify the core need as structure-aware anomaly amplification. Therefore, we design a dual-channel heterogeneous graph capsule network. Unlike standard graph based methods, its dynamic routing mechanism iteratively reinforces coupling coefficients for anomalous feature pathways, effectively preserving and highlighting localized deceptive patterns that are otherwise lost during message passing.

Decision-Oriented Alignment (Addressing the “How consistent should it be?” gap). Traditional domain alignment methods like Maximum Mean Discrepancy (MMD) apply a uniform metric to all samples, which is fundamentally flawed for a binary classification task. To address this issue, we reformulate FND as a hierarchical domain alignment task and propose Local-Enhanced Maximum Mean Discrepancy (LE-MMD). This novel metric explicitly maximizes cross-modal divergence for fake news while minimizing it for real news, and crucially, it does so in a locality-enhanced manner to capture fine-grained inconsistencies, which transforms a static distance measure into a dynamic, task-driven optimization objective.

The synergy of these three components creates a framework that is more than the sum of its parts. Semantic enrichment provides the raw evidence, the graph capsule network preserves the structural anomalies in that evidence, and LE-MMD provides the precise, decision-oriented objective for the final classification. This integrated approach ensures sensitivity to both the global semantic context and the local structural anomalies characteristic of fake news.

The main innovations of this paper are as follows:

- **Semantic enrichment via LLM-based multi-perspective generation.** To address the limitation that existing methods rely on surface-level features and miss implicit deceptive cues, we design prompt templates that leverage LLMs to generate multi-perspective image descriptions and supplement latent external entities, constructing an augmented corpus that makes semantic misleading signals explicit for detection.
- **Anomaly-preserving representation via graph capsule network.** To overcome the issue that standard graph based methods smooth away local anomalies such as partial tampering, we map multimodal data into a dual-channel heterogeneous graph and design a graph capsule network with dynamic routing, which iteratively strengthens coupling coefficients for anomalous pathways to preserve and highlight localized deceptive patterns.
- **Decision-oriented alignment via LE-MMD.** To remedy the flaw that traditional distribution alignment applies a uniform metric to all samples regardless of their ground truth, we reformulate FND as a hierarchical domain alignment task and propose LE-MMD, which separately minimizes cross-modal divergence for real news while maximizing it for fake news with locality-aware sensitivity.
- A large number of experimental results on three benchmark FND datasets show that the method constructed in this paper outperforms the state-of-the-art method.

2. Research objectives

Our goal is to construct a multimodal FND framework that systematically decomposes cross-modal consistency into three complementary dimensions and addresses each with targeted solutions. Specifically, we pursue the following three research objectives:

RO1. Semantic enrichment via LLM-based multi-perspective generation.

To address the limitation that existing methods rely on surface-level features and miss implicit deceptive cues, we design an LLM-enhanced framework that automatically generates multi-perspective image descriptions and supplements external entities, constructing a semantically augmented corpus that makes misleading signals explicit for detection.

RO2. Anomaly-preserving representation learning via graph capsule network.

To overcome the issue that standard graph based methods smooth away local anomalies such as partial tampering, we develop a graph based representation learning framework that maps multimodal data into a dual-channel heterogeneous graph and employs a graph capsule network with dynamic routing, which iteratively strengthens coupling coefficients for anomalous pathways to preserve and highlight localized deceptive patterns.

RO3. Decision-oriented cross-modal alignment via LE-MMD.

To remedy the flaw that traditional distribution alignment applies a uniform metric to all samples regardless of their ground truth, we propose a hierarchical domain alignment strategy that explicitly splits news data into real and fake domains and applies Local-Enhanced Maximum Mean Discrepancy to separately align cross-modal distributions—minimizing divergence for authentic content while maximizing it for deceptive content with locality-aware sensitivity.

3. Related work

3.1. Unimodal fake news detection

Unimodal FND methods mainly identify fake information by analyzing abnormal patterns in a single modal such as text or images. Park et al. (2025) proposed an adversarial style enhancement method aimed at training a fake news detector that can resist various style conversion attacks. The core mechanism lied in strategically utilizing LLM to automatically generate diverse and

coherent text style conversion attack prompts, particularly enhancing the generation of prompts that pose extreme challenges to the detector. [Chen et al. \(2025\)](#) developed a FND model based on news document, using LLM to generate external knowledge as supplementary information only for the training set, and introduced a graph based semantic perception feature alignment module to solve knowledge contradictions, as well as an information bottleneck based knowledge distillation module to ensure implicit generation of these features during the inference process. [Hu et al. \(2024\)](#) designed an Adaptive Basis Guided Network (ARG) to enable SLM to selectively absorb news analysis insights provided by LLM. And further derived a lightweight version of ARG-D through distillation technology that did not require real-time calling of LLM. [Karaoglan \(2024\)](#) proposed a FND method based on attention mechanism deep multi instance learning, which utilized ensemble attention mechanism to adaptively focus on key instances and dynamically adjusted attention weights, thus fully leveraging the learning ability of Multiple Instance Learning model. To detect fake news through user comments, [Nan et al. \(2024\)](#) used LLMs as user simulators and comment generators, and designed a feedback enhanced detection system that generated comments by inputting diverse user profiles into LLMs and aggregating comments generated from different subgroups. [Rathinapriya and Kalaivani \(2025\)](#) proposed a novel deep learning approach specifically designed for identifying fake news in epidemic situations. Firstly, text data was collected from benchmark resources related to the epidemic. Then, Glove, Bert, and TF-IDF were fused and optimized with a weighted feature selection module to extract features. Finally, the optimal attention mechanism was used to classify fake news using a multi-scale dense network combined with LSTM layers.

3.2. Multimodal fake news detection

However, text detection is susceptible to semantic confusion and contextual loss, and image detection is difficult to recognize advanced forgery techniques and cannot capture cross modal contradictions. To address the above issues, Multimodal Fake News Detection (MFND) methods identify fake content by jointly analyzing the correlation and consistency between different modalities such as text, images, and videos, and mining potential contradictions or complementary information. [Bazmi et al. \(2024\)](#) proposed a novel uncertainty aware decoupling representation learning framework. Firstly, a probabilistic representation module is used to model multimodal information as Gaussian distribution. Secondly, a decoupling representation learning framework is introduced to separate modal shared information and private information. Finally, a dynamic fusion module based on uncertainty can adjust modal weights according to confidence. [Gedara et al. \(2025\)](#) proposed a novel MFND framework based on fuzzy logic, which integrated text, images, and their semantic correlations to obtain more accurate and interpretable results, which included a LSTM network for text analysis, an adaptive neural fuzzy inference system for image classification, a similarity analysis module for detecting image text consistency, and a fuzzy inference system for deriving final conclusions. [Luvembe et al. \(2025\)](#) proposed an adaptive automatic fusion framework based on hierarchical attention for MFND, which directly integrated image descriptions into the fusion process to enhance cross modal learning. [Jiang et al. \(2025\)](#) proposed a MFND model that enhanced multimodal features through single modal features. By transforming the classification of n samples into a more robust $(n \times z)$ sample problem, the performance of small sample MFND can be significantly improved. [Liu et al. \(2025\)](#) designed a bidirectional multi-scale fine-grained interaction mechanism to ensure deep collaboration between entities and image blocks. A multi-stream attention interaction mechanism was introduced to enhance the interaction between text and images. In addition, a dynamic routing fusion mechanism was adopted to dynamically fuse fine-grained features and coarse-grained features.

3.3. LLM based fake news detection

Although MFND methods have made some progress, they still struggle to effectively model complex semantic relationships across modalities, and the interpretability of the detection process remains limited. To address these issues, LLM based methods leverage the powerful semantic understanding and reasoning capabilities of LLMs, demonstrating significant advantages in areas such as multimodal fusion, dynamic knowledge integration, and interpretability enhancement. [Wu et al. \(2025\)](#) proposed a balanced cross modal prompt learning method, designed a hierarchical cross modal prompt interaction module, proposed a modal balance learning module, and designed an event robustness module. [Hu et al. \(2025\)](#) proposed a multimodal detection framework that integrates LLM with global label propagation. It constructs a cross-modal correlation graph by extracting deep semantic features from the large language model and uses the label propagation mechanism to enhance the modeling ability for complex multimodal relationships. [Jin et al. \(2025\)](#) designed a detection model driven by dynamic knowledge update. This model continuously integrates external knowledge bases through the large language model to achieve dynamic adaptation and iterative optimization of the detection capability for the evolution patterns of false information. [Zheng et al. \(2025\)](#) developed an inference-enhanced detection method based on large visual language models. It generates interpretable analysis bases while predicting the authenticity of news and enhances the transparency and credibility of the detection process through multi-modal reasoning paths. [Jiang and Wang \(2025\)](#) constructed a context-aware multimodal detection system, using the large visual language model to capture the contextual inconsistency between text and visual content under a unified prompt framework. [Yang et al. \(2025\)](#) conducted a systematic benchmark study of large language models in anomaly detection tasks. Through the construction of a multi-dimensional evaluation system, it deeply analyzed the boundary and improvement direction of LLM's ability to identify covert deception patterns in news content. These studies jointly promoted the development of false news detection technology based on large language models from multiple dimensions such as feature fusion, knowledge update, interpretability enhancement, and system evaluation. [Jin, Wang et al. \(2025\)](#) presented a comprehensive taxonomy of AI-driven automation technologies for misinformation detection, systematically categorizing existing approaches into knowledge-driven, data-driven, and hybrid paradigms while highlighting the evolving role

Table 1
Symbols and their meanings.

Symbol	Meaning	Symbol	Meaning
D_i	Document of the i th news	P_i	Image of the i th news
D_i^V	Feature vector set of news document	P_i^V	Feature vector set of news image
d_{ik}^V	Feature vector of the k th token in document	p_{ik}^V	Feature vector of the k th patch in image
ED_i^j	j th enhanced document for the i th news	VED_i^j	Feature vector set of enhanced document
ved_{it}^j	Feature vector of the t th token in enhanced document	MED_i^j	Mean vector of enhanced document
$\overline{MED}_i$	Centroid vector of all enhanced descriptions	sim_i^j	Cosine similarity between description and centroid
τ	Quality control threshold	ID_i	Set of initial external entities
id_i^j	j th initial external entity	SD_i	Set of supplementary external entities
sd_i^j	j th supplementary external entity	F_i	Set of merged external entities
f_i^j	j th external entity	vf_i^j	Feature vector of external entity
G_1	Heterogeneous graph with enhanced documents	G_2	Heterogeneous graph with external entities
V_1, V_2	Node sets	E_1, E_2	Edge sets
X_1, X_2	Node feature matrices	t_1, t_2	Total number of nodes
A_1, A_2	Adjacency matrices	$PMI(i, j)$	Pointwise mutual information between words i and j
$Vec - Dot_{ij}$	Dot product of document and enhanced document vectors	$TF - IDF_{ij}$	TF-IDF value of word j in document i
$\#W(i)$	Number of sliding windows containing word i	$\#W(i, j)$	Number of sliding windows containing both words i and j
$\#W$	Total number of sliding windows	V_m^1, V_m^2	Sets of masked nodes
m_r	Randomly initialized mask vector	G'_1, G'_2	Masked graphs
$W_{\pi(i,j)}$	Edge-type specific projection matrix	$\hat{u}_{j i}^{(r)}$	Vote from node i to node j 's capsule representation
$b_{ij}^{(r)}$	Log prior for coupling coefficient	$c_j^{(r)}$	Routing coefficient
$s_j^{(r)}$	Weighted input capsule	$v_j^{(r)}$	Output capsule
RI	Number of routing iterations	ϵ	Small constant to prevent division by zero
$D_T, D_T^{(1)}, D_T^{(2)}, P_T$	Feature sets for real news	$D_F, D_F^{(1)}, D_F^{(2)}, P_F$	Feature sets for fake news
d_T, d_F	Number of real and fake news samples	$\mu_s^{real}, \mu_s^{fake}$	Mean values of segment s for real and fake news

of LLMs in automating fact-checking pipelines. Jin, Su et al. (2025) proposed Courtroom-FND, a novel multi-role FND method that simulates courtroom debate dynamics through an argument switching mechanism. By assigning prosecution and defense roles to different LLM agents and forcing them to adopt alternative perspectives, their framework exposes hidden inconsistencies in news content through adversarial deliberation, achieving significant accuracy gains on challenging datasets. Jin, Gao et al. (2025) developed a veracity-oriented context-aware prompting optimization framework for FND that integrates background enrichment, analogical reasoning, and self-adaptive bootstrap optimization to steer LLMs toward truthful interpretations while mitigating hallucinations.

Although many researchers have conducted extensive FND studies, existing methods, including traditional deep learning approaches, LLM-based reasoning frameworks such as Courtroom-FND's adversarial debate, and veracity-constrained prompting optimization like CAPE-FND, exhibit fundamental limitations in detecting advanced deception strategies such as partial tampering and semantic misleading. These approaches either operate on surface-level features that miss implicit cues, employ aggregation mechanisms that smooth away local anomalies, or apply uniform alignment metrics that fail to distinguish between the distinct consistency patterns of real and fake news. To address these limitations, we propose LGCNLA, a novel framework that systematically targets three critical dimensions of deception. At the semantic layer, we leverage LLMs to generate multi-perspective image descriptions and supplement external entities, transforming implicit misleading cues into explicit evidence. At the structural layer, we construct a dual-channel heterogeneous graph capsule network whose dynamic routing mechanism preserves and amplifies local anomalous patterns that traditional graph-based methods would smooth away. At the distributional layer, we introduce LE-MMD to separately align cross-modal distributions for real and fake news, minimizing divergence for authentic content while maximizing it for deceptive content with locality-aware sensitivity.

4. Methodology

LLM-Guided Graph Capsule Network with Local-Enhanced Alignment (LGCNLA) based FND model is constructed in this paper, and the structure is shown in Fig. 2. (1) The eigenvectors of news documents and news images are obtained based on the pre-trained models. (2) Considering the differences in readers' age, education, work and attention, the LLM based News Readers Prompt (NRP) template is constructed, and the enhanced documents for each news image are generated. (3) The external entities of the news documents are generated by the pre-trained SpaCy and the LLM based External Entity Supplementary Prompt (EESP) template. (4) The news is mapped into a dual-channel heterogeneous graph. The type-aware node selection based graph masking strategy is proposed. The graph capsule network is constructed and the eigenvectors of the news document nodes in both graphs are dynamically updated through dynamic routing mechanism and attention mechanism. (5) The FND corpus is split into the real news and fake news corpora, and LE-MMD is proposed to perform hierarchical domain alignment. To facilitate understanding, we provide explanations for the main symbols in Table 1.

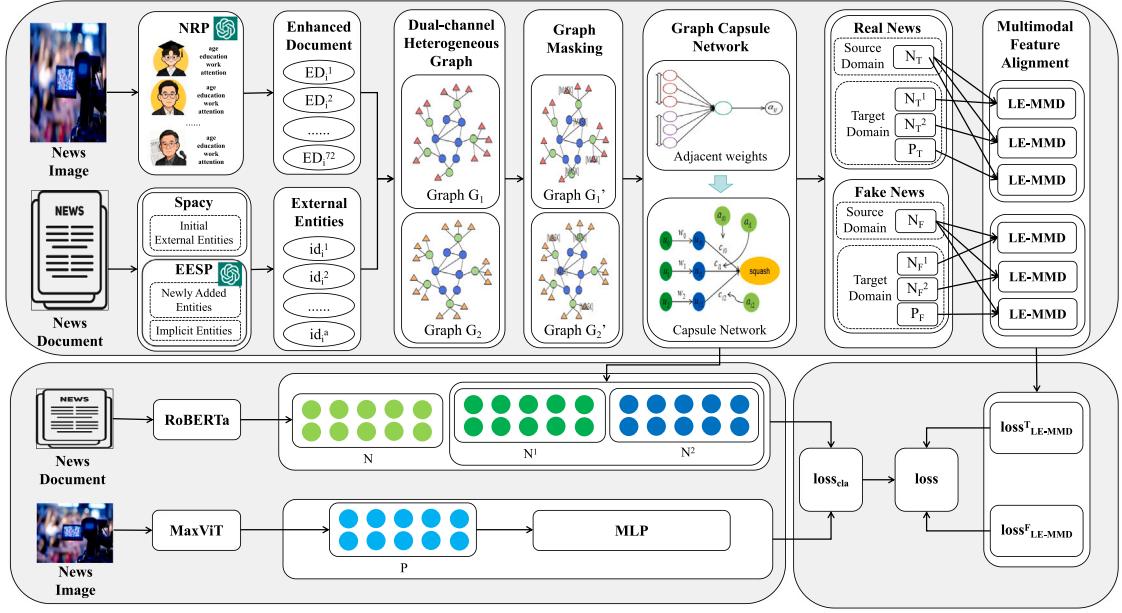


Fig. 2. The structure of LGCNLA (blue circles represent word nodes, green circles represent news document nodes, red triangles represent external entity nodes, and orange triangles represent enhanced document nodes).

4.1. Feature extraction

Given a news corpus $C = \{\{D_1, P_1\}, \{D_2, P_2\}, \dots, \{D_n, P_n\}\}$ that contains news documents and images, where n represents the number of news. For the i th ($1 \leq i \leq n$) news, the document and image are respectively transformed into eigenvectors through the pre-trained models. For document D_i , it is transformed into the document eigenvector $D_i^V \in \mathbb{R}^{x \times d_d}$ through pre-trained RoBERTa (Liu et al., 2019):

$$D_i^V = \text{RoBERTa}(D_i) = \{d_{i1}^V, d_{i2}^V, \dots, d_{ix}^V\} \quad (1)$$

Where x represents the number of tokens in the document, and d_d represents the hidden dimension of the text representation.

For image P_i , it is transformed into the image eigenvector $P_i^V \in \mathbb{R}^{z \times d_p}$ through pre-trained MaxViT (Tu et al., 2022):

$$P_i^V = \text{MaxViT}(P_i) = \{p_{i1}^V, p_{i2}^V, \dots, p_{iz}^V\} \quad (2)$$

Where z represents the number of image patches, and d_p represents the hidden dimension of the image representation.

4.2. LLM based enhancing multimodal representation

To address the challenge of insensitivity to advanced deceptive strategies such as partial tampering and semantic misguidance, we employ two complementary approaches to enhance multimodal representation. Firstly, semantically enriched and multi-perspective descriptions of news images are generated to amplify subtle visual anomalies including tampering traces and contextual inconsistencies, which reveals deceptive visual cues that are frequently overlooked in raw image data. Secondly, external entities are extracted from news documents, supplementing textual content with contextually relevant and implicit entities to identify manipulated or omitted references that indicate semantic distortion.

4.2.1. Acquisition of news image enhanced document

A single image can be interpreted from multiple semantic perspectives. For instance, an image accompanying fake news might be superficially relevant to the text but contain subtle visual inconsistencies (e.g., an altered logo, a mismatched background) or imply a different event when viewed with varying degrees of domain knowledge. Relying on a single, literal description of the image, as done in most existing methods, limits the model's exposure to these potentially deceptive cues.

To address this issue, we propose to leverage LLMs to generate multi-perspective image descriptions. Instead of aiming to simulate real readers which are inherently difficult to validate, we treat the combination of attributes as a controllable mechanism to sample a diverse and structured semantic space. Each combination acts as a distinct semantic lens that guides the LLM to focus on different aspects of the image. This process is designed to enrich the representation of the image by capturing a broader range of both explicit and implicit visual cues that a single description might miss.

For the i th news $\{D_i, P_i\}$, fake news can be effectively identified by reader's description of the content of image P_i and then associating it with document D_i . However, researches show that people of different age groups have significantly different understandings of the same image. In addition, differences in education and work can also lead people to give different descriptions of the same image. What is more, people who pay more attention to news usually provide more in-depth image descriptions than those who pay less attention. Therefore, considering the differences in readers' age, education, work and attention, the LLM based News Readers Prompt (NRP) template is constructed to generate the enhanced documents for each news image. The assignments for the four attributes are as follows:

Age: 18–35; 36–55; > 55

Education: High school or below; bachelor's degree; master's degree; doctoral degree

Work: Related to news; not related to news

Attention: less than 2 h per week; 2 to 8 h per week; more than 8 h per week

Given that people under the age of 18 rarely pay attention to news, only news readers who are older than 18 are considered. By randomly selecting the above four attributes, a total of 72 different combinations are obtained, and the LLM based NRP template is constructed to generate 72 enhanced documents for each news image. The NRP template is as follows:

The news readers prompt template

Suppose you are a news reader and you have the following attributes:

- Your age is: [age]
- Your educational attainment is: [education]
- Your work is: [work]
- Your level of attention to the news is: [attention]

Given a news image $[P_i]$, please provide a description of this image based on your attribute settings. Note that the narrative logic of news should be adopted for description and be as concise as possible.

After obtaining 72 enhanced documents $\{ED_i^1, ED_i^2, \dots, ED_i^{72}\}$ of the news image P_i , for each enhanced document ED_i^j ($1 \leq j \leq 72$), it is transformed into the document eigenvector $VED_i^j \in \mathbb{R}^{r \times d_d}$ through pre-trained RoBERTa:

$$VED_i^j = \text{RoBERTa}(ED_i^j) = \{ved_{i1}^j, ved_{i2}^j, \dots, ved_{ir}^j\} \quad (3)$$

Where r represents the number of tokens in document ED_i^j . Next, the mean vector MED_i^j of ED_i^j is obtained by calculating the mean of the eigenvectors of all tokens in ED_i^j :

$$MED_i^j = \text{mean} \left(\sum_{i=1}^r ved_{it}^j \right) = \frac{ved_{i1}^j + ved_{i2}^j + \dots + ved_{ir}^j}{r} \quad (4)$$

Generating multiple descriptions from varied prompts inevitably introduces the risk of noise, hallucinations, or outlier interpretations that deviate significantly from the core semantic content of the image. To mitigate this risk and ensure the reliability of the augmented corpus, we introduce a quality control mechanism based on semantic consistency filtering.

First, we compute the centroid vector $\overline{MED}_i$ that represents the central semantic tendency of all generated descriptions for image P_i :

$$\overline{MED}_i = \frac{1}{72} \sum_{j=1}^{72} MED_i^j \quad (5)$$

Next, for each enhanced document ED_i^j , we calculate the cosine similarity between its mean vector MED_i^j and the centroid vector $\overline{MED}_i$:

$$sim_i^j = \frac{MED_i^j \cdot \overline{MED}_i}{\|MED_i^j\| \|\overline{MED}_i\|} \quad (6)$$

The cosine similarity measures the semantic alignment between an individual description and the collective semantic center of all descriptions. A low similarity score indicates that the corresponding description deviates substantially from the mainstream interpretation of the image, potentially containing hallucinations, exaggerated speculations, or irrelevant content. To filter out such low-quality descriptions, we introduce a threshold parameter $\tau \in [0, 1]$ (the optimal value of τ will be verified in the experimental section.), and only the results where $sim_i^j > \tau$ are retained.

4.2.2. Acquisition of news document external entity

Fake news often spreads on a single platform, while true information repeatedly appears on multiple authoritative platforms such as mainstream media and official accounts. External entities provide cross-platform data to help identify abnormal propagation patterns. Therefore, by using external knowledge sources such as Wikipedia and professional databases, it is possible to verify

whether the images, events and data in the news match the existing records. SpaCy (Nivre, 2008) is a widely used entity generation tool that acquires external entities of news documents through methods such as named entity recognition models, knowledge base links and rules, and verifies their authenticity in combination with external data sources. First, the pre-trained SpaCy is selected to obtain the initial external entities of news document D_i :

$$ID_i = \{id_i^1, id_i^2, \dots, id_i^a\} \quad (7)$$

Where id_i^j ($1 \leq j \leq a$) represents the i th initial external entity of D_i , and a represents the number of initial external entities of D_i .

However, the pre-trained SpaCy relies on the data distribution during training and has difficulty identifying newly added entities in news. Moreover, SpaCy only recognizes explicit entities and has difficulty identifying implicit entities or relationships in the text. Therefore, based on the initial external entities ID_i of the news document D_i , the latest and implicit external entities are generated by constructing the LLM based External Entity Supplementary Prompt (EESP) template, which is defined as follows:

The external entity supplementary prompt template

Suppose you are a senior domain expert in the news field. Given a news document [D_i], the initial external entities [ID_i] have been extracted through the pre-trained SpaCy model. Based on this, please complete:

- Identify the newly added entities in the news document
- Identify the implicit entities in the news documents
- Classify the importance of entities into three levels: high, medium and low, and mark the importance of entities

Output example:

{ "Newly Added Entity": [{ "Entity name": ".....", "Importance": "....."; }],
"Implicit Entity": [{ "Entity name": ".....", "Importance": "....."; }] }

Since entities of low importance play little role in determining the authenticity of news, only entities of high and medium importance are retained to obtain supplementary external entities SD_i of news document D_i :

$$SD_i = \{sd_i^1, sd_i^2, \dots, sd_i^b\} \quad (8)$$

Where sd_i^j ($1 \leq j \leq b$) represents the j th supplementary external entity of D_i , and b represents the number of supplementary external entities.

Then, the initial external entities ID_i and the supplementary external entities SD_i are merged to obtain the external entities F_i :

$$F_i = ID_i \cup SD_i = \{id_i^1, id_i^2, \dots, id_i^a, sd_i^1, sd_i^2, \dots, sd_i^b\} = \{f_i^1, f_i^2, \dots, f_i^{a+b}\} \quad (9)$$

For any external entity f_i^j ($1 \leq j \leq a + b$) in F_i , the pre-trained RoBERTa is used to transform it into the eigenvector $vf_i^j \in \mathbb{R}^{d_d}$:

$$vf_i^j = RoBERTa(f_i^j) \quad (10)$$

To mitigate the risk that LLM-generated entities may perpetuate or amplify misinformation present in fake news, we introduce a verification mechanism that validates all high and medium importance entities against external knowledge sources.

Stage 1: Knowledge Base Retrieval. We query multiple trusted knowledge sources, including Wikipedia, Wikidata, and Google Knowledge Graph, to verify the existence and factual consistency of the entity. For each entity, we retrieve:

Existence verification: Does the entity have a corresponding page/article in at least two independent knowledge bases?

Factual consistency: For entities with temporal attributes (e.g., events, dates), we verify consistency with the news document's timestamp.

Stage 2: Cross-Documents Consistency Check. For entities that appear in multiple news items within our corpus, we check their contextual consistency. If the same entity is associated with contradictory attributes across different documents, it is flagged for additional scrutiny.

Only entities that pass both verification stages are retained in the final entity set.

4.3. Construction of the graph based representation learning framework

4.3.1. Construction of the dual-channel heterogeneous graph

As traditional sequence-based methods are often constrained by linear structures when processing multimodal information, which struggle to effectively extract discontinuous semantic relationships between cross-modal entities and tend to blur local anomalous features during long-range dependency modeling. Therefore, we transform the news into a dual-channel heterogeneous graph structure, which not only preserves the independent features of news documents, image descriptions, and external entities but also enables multi-hop semantic reasoning through the message-passing mechanism of GNNs. Therefore, it can accurately capture cross-modal inconsistency features and enhance the model's sensitivity to complex deceptive patterns such as partial tampering and semantic misleading.

Given the news document corpus D , the enhanced document corpus ED , and the external entities corpus FD , where u represents the remaining number of enhanced documents after the quality control mechanism. D and ED are transformed into graph $G_1 = (V_1, E_1, X_1)$, where V_1 represents the set of word nodes, news document nodes and enhanced document nodes, and E_1 represents the set of word-word edges, news document-word edges and news document-enhanced document edges, and X_1 represents the set of node features. D and FD are transformed into graph $G_2 = (V_2, E_2, X_2)$, where V_2 represents the set of word nodes, news document nodes, and external entity nodes, and E_2 represents the set of word-word edges, news document-word edges and news document-external entity edges, and X_2 represents the set of node features. Suppose the total number of nodes $|V_1| = t_1$ and $|V_2| = t_2$. Let $A_1 \in \mathbb{R}^{t_1 \times t_1}$ represents the adjacency matrix of G_1 , $A_2 \in \mathbb{R}^{t_2 \times t_2}$ represents the adjacency matrix of G_2 , and $A_{num}(i, j)$ ($num = 1, 2$) represents the element in the i th row and j th column. Then:

$$A_1(i, j) = \begin{cases} PMI(i, j) & i, j \text{ are words, } PMI(i, j) > 0 \\ Vec - Dot_{ij} & i \text{ is news document, } j \text{ is enhanced document} \\ TF - IDF_{ij} & i \text{ is news document, } j \text{ is word} \\ 1 & i = j \\ 0 & \text{otherwise} \end{cases} \quad (11)$$

$$A_2(i, j) = \begin{cases} PMI(i, j) & i, j \text{ are words, } PMI(i, j) > 0 \\ cos(i, j) & i \text{ is news document, } j \text{ is external entity, } cos(i, j) > 0.3 \\ TF - IDF_{ij} & i \text{ is news document, } j \text{ is word} \\ 1 & i = j \\ 0 & \text{otherwise} \end{cases} \quad (12)$$

Let the news document vector be the mean of all word vectors in the document with dimension $d_{doc} = 768$, let the enhanced document vector with dimension $d_{doc} = 768$, and $Vec - Dot_{ij}$ represents the dot product operation of them. Due to the need to determine whether a news document is fake news through all enhanced documents, therefore, all dot product results are retained. $cos(\cdot, \cdot)$ represents the cosine similarity between the news document vector (dimension $d_{doc} = 768$) and the external entity vector (dimension $d_{ext} = 768$), and only results greater than 0.3 are retained. This threshold is determined through empirical observation on a validation set. We experiment with values ranging from 0.1 to 0.5 in increments of 0.05 and find that 0.3 achieves the optimal balance between retaining sufficient document-entity connections and filtering out noisy, semantically irrelevant associations. $TF - IDF$ represents the term frequency-inverse document frequency of the word vector in the document vector, and IDF is calculated based on the entire corpus. The PMI value of a word pair i, j is computed as follows:

$$PMI(i, j) = \log \frac{p(i, j)}{p(i)p(j)} \quad (13)$$

$$p(i, j) = \frac{\#W(i, j)}{\#W} \quad (14)$$

$$p(i) = \frac{\#W(i)}{\#W} \quad (15)$$

Where $\#W(i)$ represents the number of sliding windows that contain word i , $\#W(i, j)$ represents the number of sliding windows that contain both word i and j , and $\#W$ represents the total number of sliding windows in the corpus. The number of sliding windows is set to 5 after numerous experiments. A positive PMI indicates high semantic correlation, while a negative PMI indicates little or no correlation. Therefore, we only add edges between word pairs with positive PMI values.

4.3.2. Type-aware node selection based graph masking strategy

Inspired by the masked language modeling objective in BERT (Devlin et al., 2019), which randomly masks a portion of input tokens to learn robust contextualized representations, we introduce a type-aware graph masking strategy to enhance the robustness and generalization of our graph capsule network. This approach serves two purposes. (1) It acts as a regularization mechanism that prevents the model from overfitting to potentially noisy or misleading nodes (e.g., LLM-generated enhanced descriptions that may contain hallucinations). (2) It encourages the dynamic routing mechanism to learn more context-aware representations by predicting or compensating for missing information.

Following BERT's design principle where different token types implicitly receive different treatment, we assign type-specific masking ratios to account for the varying reliability and information density of different node types in our heterogeneous graphs. Enhanced document nodes in G_1 and external entity nodes in G_2 are generated by LLMs and may contain hallucinations or biases, therefore, they are assigned a higher masking ratio (50%) to encourage the model not to over-rely on potentially noisy signals. Word nodes, which are numerous and statistically robust across the corpus, receive a lower ratio (20%). News document nodes, as the core entities for classification, receive an intermediate ratio (30%) to balance regularization with information preservation. Let V_M^1 and V_M^2 represent the masking node sets of G_1 and G_2 respectively. For any node that needs to be masked, we replace it with a randomly initialized vector m_r . Therefore, the node features X_1 and X_2 of G_1 and G_2 are respectively transformed into $\hat{X}_1$ and $\hat{X}_2$ as follows.

$$\hat{X}_1^i = \begin{cases} X_1^i & \text{if } V_i \notin V_M^1 \\ m_r & \text{if } V_i \in V_M^1 \end{cases} \quad (16)$$

$$\hat{X}_2^j = \begin{cases} X_2^j & \text{if } V_j \notin V_M^2 \\ m_r & \text{if } V_j \in V_M^2 \end{cases} \quad (17)$$

Where X_i^i represents the i th node feature of G_1 and X_j^j represents the j th node feature of G_2 . Therefore, G_1 and G_2 are respectively transformed into $G'_1 = (V_1, E_1, \hat{X}_1)$ and $G'_2 = (V_2, E_2, \hat{X}_2)$.

4.3.3. Graph capsule network

Traditional graph based methods exhibit significant limitations when processing heterogeneous graph structures for FND. Their feature aggregation mechanism, typically based on neighborhood averaging, acts as a low-pass filter that progressively smooths node representations during multi-layer propagation. While this smoothing effect captures global consensus, it inadvertently dilutes local anomalous features, which indicate partial tampering or localized semantic inconsistencies. Furthermore, standard GNNs struggle to explicitly model part-whole hierarchical relationships, treating all features as equally contributing to the final representation and thus failing to distinguish between globally consistent patterns and locally anomalous cues. To address these issues, we construct a graph capsule network with a dynamic routing mechanism. Unlike conventional aggregation that averages neighbor information, capsule networks preserve feature hierarchies by representing each node as a vector capsule whose orientation encodes semantic properties. The dynamic routing mechanism iteratively refines coupling coefficients between capsules through agreement-based updates, strengthening pathways that exhibit mutually supportive patterns while preserving isolated anomalies as distinct capsules if they represent meaningful deceptive signals. This creates a contrast enhancement effect that amplifies diagnostically valuable local patterns while suppressing noise. With type-specific projection matrices for different edge categories and cosine similarity-based agreement updates to ensure gradient stability, the graph capsule network not only preserves structural information but also enhances detection robustness against advanced deceptive strategies such as partial tampering and semantic misleading.

We construct a graph capsule network that operates on the masked heterogeneous graphs G'_1 and G'_2 . Let $G' = (V, E, \hat{A}, \hat{X})$ represent either graph, where $\hat{A}$ is the normalized adjacency matrix from Section 4.3.1, and $\hat{X}$ is the node feature matrix from Section 4.3.2. The degree matrix M_D of graph G' is calculated based on $\hat{A}$, and the normalized symmetric adjacency matrix $\tilde{A} = M_D^{-\frac{1}{2}} \hat{A} M_D^{-\frac{1}{2}}$. The graph capsule network consists of L capsule layers, each performing a dynamic routing process that preserves and amplifies local anomalous patterns.

For each edge $(i, j) \in E$, we define a type-specific projection matrix $W_{\pi(i,j)}$, where $\pi(i, j)$ denotes the edge type. This allows different semantic relationships to be modeled with different parameterizations. The projected feature from node i to node j is:

$$\hat{u}_{j|i} = \tilde{A}_{ij} \cdot W_{\pi(i,j)} h_i \quad (18)$$

Where $\hat{u}_{j|i}$ represents the vote from node i to node j 's capsule representation, $\tilde{A}_{ij}$ represents the value of the i th row and j th column in matrix $\tilde{A}$, h_i represents the input eigenvector of node i .

Dynamic Routing Mechanism. Dynamic routing iteratively refines coupling coefficients to strengthen consistent pathways. Unlike GNNs that average neighbor features, capsule routing amplifies mutually supportive features and suppresses inconsistent ones, which are crucial for detecting local anomalies.

Let $b_{ij}^{(0)}$ be the initial log prior that node i should be coupled to node j . We initialize $b_{ij}^{(0)} = 0$ for all edges, treating all neighbors equally at the start. In the following content, ri represents the routing iteration from 1 to $RI = 3$. The coupling coefficients $c_{ij}^{(ri)}$ are updated iteratively based on agreement between the predicted capsule $\hat{u}_{j|i}$ and the output capsule $v_j^{(ri)}$ from the previous iteration.

Step 1: Compute routing coefficients.

$$c_{ij}^{(ri)} = \frac{\exp(b_{ij}^{(ri-1)})}{\sum_{k \in N(j)} \exp(b_{ik}^{(ri-1)})} \quad (19)$$

Where $N(j)$ is the set of neighbors of node j . The coefficients $c_{ij}^{(ri)}$ sum to 1 over all neighbors of node j , ensuring a proper distribution.

Step 2: Compute weighted input capsule.

$$s_j^{(ri)} = \sum_{i \in N(j)} c_{ij}^{(ri)} \hat{u}_{j|i} \quad (20)$$

Where weighted input capsule $s_j^{(ri)}$ aggregates all neighbor votes weighted by $c_{ij}^{(ri)}$.

Step 3: Apply squash activation.

$$v_j^{(ri)} = \text{squash}(s_j^{(ri)}) = \frac{\|s_j^{(ri)}\|^2}{1 + \|s_j^{(ri)}\|^2} \frac{s_j^{(ri)}}{\|s_j^{(ri)}\|} \quad (21)$$

The squash function preserves the direction of the vector while normalizing its length to between 0 and 1, with longer vectors being preserved and shorter vectors being suppressed. The final output capsule after RI iterations is $v_j = v_j^{RI}$.

Step 4: Update log priors (agreement step).

$$b_{ij}^{(ri)} = b_{ij}^{(ri-1)} + \frac{v_j^{(ri)} \cdot \hat{u}_{j|i}}{\|v_j^{(ri)}\| \|\hat{u}_{j|i}\| + \epsilon} \quad (22)$$

Where ϵ represents division by zero. The agreement is measured by the cosine similarity between the output capsule $v_j^{(ri)}$ and the input vote $\hat{u}_{j|i}$, ensuring that the update is scale-invariant and stable.

After RI iterations, we obtain the final output capsules v_j . This routing process amplifies features that are consistent across multiple neighbors, as their agreements reinforce the corresponding coupling coefficients, while inconsistent features receive lower coefficients and are effectively suppressed.

Attention-Augmented Routing. While dynamic routing captures feature-level agreements, we also incorporate a content-based attention mechanism to focus on semantically relevant neighbors. We compute attention scores based on node features:

$$e_{ij} = \text{LeakyReLU} \left(a^T [h_i \parallel h_j] \right) \quad (23)$$

$$a_{ij} = \frac{\tilde{A}_{ij} \exp(e_{ij})}{\sum_{k \in N(j)} \tilde{A}_{ik} \exp(e_{ik})} \quad (24)$$

Where T represents transposition and $[\cdot \parallel \cdot]$ represents the concatenation operation. The final output capsule is computed as a convex combination of the routing-based and attention-based aggregations:

$$s'_j = \sum_{i \in N(j)} (\lambda c_{ij} + (1 - \lambda) a_{ij}) \hat{u}_{j|i} \quad (25)$$

$$v_j^{\text{final}} = \text{squash} \left(s'_j \right) \quad (26)$$

Where $0 < \lambda < 1$ represents the hyperparameter balancing the two mechanisms. This hybrid approach combines the anomaly-amplifying property of dynamic routing with the feature-based attention, ensuring sensitivity to both structural and semantic patterns.

4.4. Multimodal feature alignment

The eigenvector $D = \{D_1^V, D_2^V, \dots, D_n^V\}$ of the news documents and the eigenvector $P = \{P_1^V, P_2^V, \dots, P_n^V\}$ of the news images are obtained in Section 4.1. Graph G_1 containing word nodes, news document nodes and enhanced document nodes, as well as graph G_2 containing word nodes, news document nodes and external entity nodes are constructed in Section 4.3.3, and the document node vector sets $D^1 = \{D_1^1, D_2^1, \dots, D_n^1\}$ of G_1 and $D^2 = \{D_1^2, D_2^2, \dots, D_n^2\}$ of G_2 are obtained by constructing the graph capsule network with an attention mechanism, where D_j^1 and D_j^2 respectively represent the eigenvectors of the j th ($1 \leq j \leq n$) node of D^1 and D^2 .

The goal of the FND task is to minimize the distance between the text and image eigenvectors of real news, while maximizing the distance between the text and image eigenvectors of fake news. To effectively measure the distribution divergence between textual and visual modalities, Maximum Mean Discrepancy (MMD) (Gretton et al., 2012) is adopted due to its capability of capturing global feature differences without requiring parametric assumptions. However, MMD primarily focuses on holistic distribution alignment and often overlooks local semantic inconsistencies, such as partial image tampering or subtle textual contradictions, which are critical for detecting sophisticated deception strategies. To address this limitation, we introduce the Multi-Granularity Local-Aware Kernel Function, which enhances MMD by segmenting features into semantically meaningful local blocks and adaptively weighting their contributions. Therefore, we innovatively propose Local-Enhanced Maximum Mean Discrepancy (LE-MMD), which preserves global alignment while reinforcing sensitivity to fine-grained cross-modal discrepancies, thereby improving the detection of localized deceptive patterns. It is important to note that LE-MMD serves exclusively as a regularization constraint during the training phase, encouraging the model to learn discriminative feature representations that capture domain-specific alignment patterns. During inference, LE-MMD is completely removed, and classification relies solely on the optimized feature extractors and the trained classification head, ensuring efficient and straightforward prediction without additional computational overhead.

After obtaining the news document eigenvector set D , the document node vector set D^1 of G_1 , the document node vector set D^2 of G_2 , and the image eigenvector P , they are split into the real news corpus $\{D_T, D_T^1, D_T^2, P_T\}$ and the fake news corpus $\{D_F, D_F^1, D_F^2, P_F\}$ according to their labels $y \in \{0, 1\}$. Then, D_T and D_F are respectively regarded as the source domain, while D_T^1, D_T^2, P_T and D_F^1, D_F^2, P_F are respectively regarded as the target domain. The smaller the distance between D_T and D_T^1, D_T^2, P_T , and the larger the distance between D_F and D_F^1, D_F^2, P_F , the more superior the model is.

4.4.1. Standard MMD revisited

For two distributions p and q with samples $X = \{x_1, x_2, \dots, x_m\}$ and $Y = \{y_1, y_2, \dots, y_l\}$, the unbiased empirical estimate of MMD is:

$$\widehat{MMD}^2(X, Y) = \frac{1}{m(m-1)} \sum_{i \neq j} k(x_i, x_j) + \frac{1}{l(l-1)} \sum_{i \neq j} k(y_i, y_j) - \frac{2}{ml} \sum_{i=1}^m \sum_{j=1}^l k(x_i, y_j) \quad (27)$$

Where $k(\cdot, \cdot)$ is a characteristic kernel. This formulation explicitly separates within-distribution similarities (first two terms) and cross-distribution similarities (third term).

4.4.2. Local-enhanced MMD

To inject locality awareness, we first partition each feature vector into S semantically meaningful segments. For a feature vector x , its s th $s = (1, 2, \dots, S)$ segment is:

$$x^s = x \left[\frac{d}{S}(s-1) : \frac{d}{S}s \right] \quad (s = 1, 2, \dots, S) \quad (28)$$

We then define a locality-aware kernel that operates on these segments:

$$k_{local}(x, y) = \sum_{s=1}^S \beta_s \cdot k(x^s, y^s) \quad (29)$$

Where β_s are segment-specific weights that quantify the diagnostic value of each local region for deception detection. Unlike variance-based weighting that may amplify noise, we define β_s based on the discriminative power of each segment, estimated using class separation statistics:

$$\beta_s = \frac{|\mu_s^{real} - \mu_s^{fake}|}{\sqrt{\sigma_s^2 + \epsilon}} \quad (30)$$

Where μ_s^{real} and μ_s^{fake} are the mean values of segment s across real and fake news samples respectively in a validation set, σ_s^2 is the pooled variance, and $\epsilon = 1e-8$ ensures numerical stability. The weights are normalized to sum to 1: $\sum_{s=1}^S \beta_s = 1$.

The final LE-MMD kernel combines global and local perspectives:

$$k_{LE}(x, y) = k_{global}(x, y) + k_{local}(x, y) \quad (31)$$

Where $k_{global}(x, y) = \exp(-\|x - y\|^2 / 2\sigma^2)$ is the standard RBF kernel.

This formulation offers several statistical advantages. (1) It maintains the unbiased estimation property of standard MMD when substituted into the three-term estimator. (2) The discriminative weights β_s naturally focus on segments with genuine deceptive signals while suppressing noise, as segments with high variance are only emphasized if they also exhibit strong class separation. (3) The normalized weights and inclusion of ϵ ensure numerical stability. (4) The resulting weights provide interpretable insights into which local feature regions are most indicative of deception. By integrating local discriminative information into a statistically sound MMD framework, LE-MMD enables precise control over cross-modal alignment at both global and local scales.

4.4.3. LE-MMD loss for hierarchical domain alignment

Substituting k_{LE} into the unbiased MMD estimator, we obtain the LE-MMD loss for real news:

$$loss_{LE-MMD}^{real} = \frac{1}{d_T(d_T-1)} \sum_{i \neq j} k_{LE}(d_T^i, d_T^j) + \frac{1}{d_T(d_T-1)} \sum_{i \neq j} k_{LE}(p_T^i, p_T^j) - \frac{2}{d_T^2} \sum_{i=1}^{d_T} \sum_{j=1}^{d_T} k_{LE}(d_T^i, p_T^j) \quad (32)$$

Where d_T represents the number of real news samples, d_T^i represents the combined textual features for the i th real news sample, and p_T^i represents the corresponding image feature.

For fake news, we aim to maximize the discrepancy. However, directly optimizing a maximization objective poses potential training risks, where it may encourage the model to learn a shortcut of simply amplifying cross-modal differences for any sample predicted as fake, potentially amplifying errors on noisy or mislabeled samples and destabilizing training. To mitigate these risks, we introduce a margin-based formulation with gradient stabilization:

$$loss_{LE-MMD}^{fake} = \max\left(0, \gamma - \widehat{MMD}_{LE}^2(D_F, P_F)\right) \quad (33)$$

Where $\gamma > 0$ is a hyperparameter that specifies the desired minimum discrepancy for fake news, and $\widehat{MMD}_{LE}^2$ is the LE-MMD estimate computed using k_{LE} .

This margin-based formulation offers several advantages over the naive negative loss. It provides bounded optimization by requiring the discrepancy to exceed a margin rather than pushing it arbitrarily large, which prevents the model from excessively amplifying differences on easy samples. In addition, it enhances noise robustness as samples already above the margin receive zero gradient, reducing the influence of outliers and mislabeled samples. What is more, it supports gradual learning through optional margin warm-up to avoid destabilizing early updates.

4.5. Fake news detection loss

After obtaining the news document feature vector set D , the image feature vector set P , the document node vector set D^1 of G_1 , and the document node vector set D^2 of G_2 , the features are transformed into the same dimension and merged. Then, the softmax function is used for classification:

$$\hat{Y} = \theta(W_o \text{relu}(W_f(\text{ave}(D, D^1, D^2) \oplus MLP_2(P)))) \quad (34)$$

Where $\theta()$ represents the softmax function, W_o and W_f both represent weight vectors, $\text{ave}(D, D^1, D^2) = \frac{MLP_1(D) + MLP_3(D^1) + MLP_3(D^2)}{3}$, MLP_1 , MLP_2 , MLP_3 and MLP_4 both represent fully connected networks, which are used to convert

Table 2
Statistical results of the three benchmark datasets.

Dataset	Fake news	Real news	Total news
Weibo	3857	3129	6986
PolitiFact	136	142	278
GossipCop	1642	4517	6159

D , D^1 , D^2 and P into the same dimension, and $\oplus$ represents vector concatenation. After obtaining the classification results, the cross-entropy loss is used to calculate the classification loss.

$$loss_{cla} = - \sum_{j=1}^t \sum_{x_i \in C} (y_{ij} \log(\hat{y}_{ij}) + (1 - y_{ij}) \log(1 - \hat{y}_{ij})) \quad (35)$$

Where C represents the entire FND corpus, $y_{ij} \in [0, 1]$ represents the prediction result, $\hat{y}_{ij}$ represents the actual label, $\hat{y}_{ij} = 0$ or $\hat{y}_{ij} = 1$. Therefore, the loss of the model is calculated as follows:

$$loss = loss_{cla} + \varphi loss_{LE-MMD}^{real} + \eta loss_{LE-MMD}^{fake} \quad (36)$$

Where $0 < \varphi, \eta < 1$ represents the weight coefficient, and the goal of model training is to minimize the loss value $loss$.

5. Experiment

5.1. Dataset

Three widely used benchmark datasets, Weibo (Jin et al., 2017), PolitiFact (Shu et al., 2017) and GossipCop (Shu et al., 2017), are selected as the datasets for this paper. Weibo is a Chinese social media dataset containing document, images and videos, labeled with real and fake news from 2012 to 2016, and is suitable for studying the spread of fake information in social networks. PolitiFact is an English political fact checking dataset from 2002 to 2018, commonly used for research on detecting fake political news. GossipCop is an English entertainment news dataset that focuses on celebrity gossip and entertainment reporting, providing multimodal content for detecting fake news in the entertainment industry. The news containing both documents and images in the three benchmark datasets are filtered, and the statistical results are shown in Table 2.

To ensure fair evaluation and prevent information leakage, we adopt different splitting strategies tailored to each dataset's characteristics. For Weibo, which consists of independent social media posts with minimal event repetition, we perform random splitting at the news item level with a ratio of 70% for training, 10% for validation, and 20% for testing. For PolitiFact and GossipCop, which contain multiple news reports covering the same events or celebrities, random item-level splitting would risk placing different expressions of the same event into both training and test sets, potentially inflating performance. Following the event-aware splitting protocols established in prior work (Wang et al., 2023), we group news items by their associated events (for PolitiFact) or celebrity subjects (for GossipCop) and perform splitting at the event/entity level, ensuring that all reports concerning the same event or celebrity appear exclusively in one partition. This prevents the model from leveraging cross-report consistency as a shortcut during evaluation.

5.2. Evaluation metrics

Accuracy, Precision, Recall and F1 are selected as evaluation metrics, which are defined as follows:

$$Accuracy = \frac{TP + TN}{P + N} \quad (37)$$

$$Precision = \frac{TP}{TP + FP} \quad (38)$$

$$Recall = \frac{TP}{TP + FN} \quad (39)$$

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (40)$$

Where P and N respectively represent the number of real and fake news, TP and TN respectively represent the number of correctly predicted real and fake news, FP and FN respectively represent the number of incorrectly predicted real and fake news.

5.3. Baseline methods

Traditional deep learning based methods MPL (W. Hu et al., 2024), QMFND (Qu et al., 2024), MoPeD (Wang et al., 2024), CSFND (Peng et al., 2024), UDRL (Li et al., 2025) and AAFHA (Luvembe et al., 2025) and LLM based methods GPT-4 (Achiam et al., 2023), AMPLE (Xu et al., 2025), AdStyle (Park et al., 2025), BCMPL (Wu et al., 2025), EARAM (Zheng et al., 2025) and AD-LLM (Yang et al., 2025) are selected as baseline methods.

Traditional deep learning based methods:

MPL: This model proposes a prompt-based learning framework that adapts pre-trained language models to FND by injecting task-specific prompts into multiple modalities.

QMFND: This model leverages quantum-inspired neural networks to fuse heterogeneous social media data, mapping high-dimensional features to quantum state representations for improved fake news classification.

MoPeD: This model identifies discriminative modality-specific factors through contrastive perception learning, then fuses them via dynamic gating for detection.

CSFND: This model constructs contextual semantic graphs to capture fine-grained incongruities between modalities, addressing cases where fake news exhibits local semantic inconsistencies.

UDRL: This model disentangles content and style representations across modalities while quantifying uncertainty, enabling detection of subtle cross-modal inconsistencies indicative of falsified information.

AAFHA: This model hierarchically fuses modalities through local-level and global-level attentions for fake news classification.

LLM based methods:

GPT-4: This model is a multimodal LLM evaluated in a zero-shot setting without any task-specific fine-tuning on the target datasets. It integrates text and image inputs through cross-modal attention mechanisms for generalized reasoning tasks, serving as a baseline to assess how well general-purpose LLMs can perform FND without supervised adaptation.

AMPLE: This model enhances prompt learning with emotion-aware attention mechanisms, jointly analyzing textual sentiment and visual affective cues to detect inconsistencies in multimodal fake news content.

AdStyle: This model generates adversarial style-transferred samples using LLMs to augment training data and improve model robustness against evolving deception tactics.

BCMPL: This model jointly optimizes prompt tuning and cross-modal fusion under a balanced learning objective, preventing dominance by any single modality during fake news classification.

EARAM: This model integrates large vision-language models to not only predict the authenticity of multimodal news but also generate interpretable analytical rationales. It enhances detection transparency by explicitly modeling the alignment between textual claims and visual evidence through cross-modal reasoning paths.

AD-LLM: This model establishes a comprehensive benchmark for evaluating LLMs in anomaly detection tasks. It systematically assesses the capability of LLMs to identify subtle deceptive patterns in news content, providing insights into their strengths and limitations for downstream fake-news detection.

5.4. Implementation details

5.4.1. Experimental environment

All experiments are conducted on four A800 GPU servers. To ensure the consistency of news documents, the method of Khattar et al. (2019) is selected for text preprocessing. For text feature extraction, we employ language-specific pre-trained RoBERTa models. For the Chinese Weibo dataset, we use the Chinese RoBERTa-wwm-ext model (Cui et al., 2020), while for the English PolitiFact and GossipCop datasets, we use the RoBERTa-base model (Liu et al., 2019). For image feature extraction, the MaxViT model (Tu et al., 2022) pre-trained on ImageNet is applied uniformly across all datasets. For LLM-based generation, OpenAI's GPT-3.5-Turbo model is employed with temperature set to 0 and top-p set to 1 by default to ensure deterministic outputs. For external entity extraction, we employ language-specific SpaCy models. For the Chinese Weibo dataset, we use the Chinese spaCy model zh_core_web_sm,¹ while for the English PolitiFact and GossipCop datasets, we use the English spaCy model en_core_web_sm.² Due to space limitations, the complete prompt templates, API configuration details, retry strategies, and post-processing rules are provided in our GitHub (<https://github.com/sygywaityou/LGCNLA>) to facilitate full reproducibility. The number of capsules is set to 64. The token numbers of the documents in the Weibo, PolitiFact and GossipCop datasets are set to 100, 300 and 200 respectively, and the dimension of each token is set to 768. The token numbers of the images in the Weibo, PolitiFact, and GossipCop datasets are all set to 200, and the dimension of each token is set to 768. The weight coefficients $\lambda = 0.6$, $\varphi = 0.7$, $\eta = 0.5$ and $\tau = 0.6$. All baseline methods use the default parameters in the original literature.

¹ <https://spacy.io>

² <https://spacy.io>

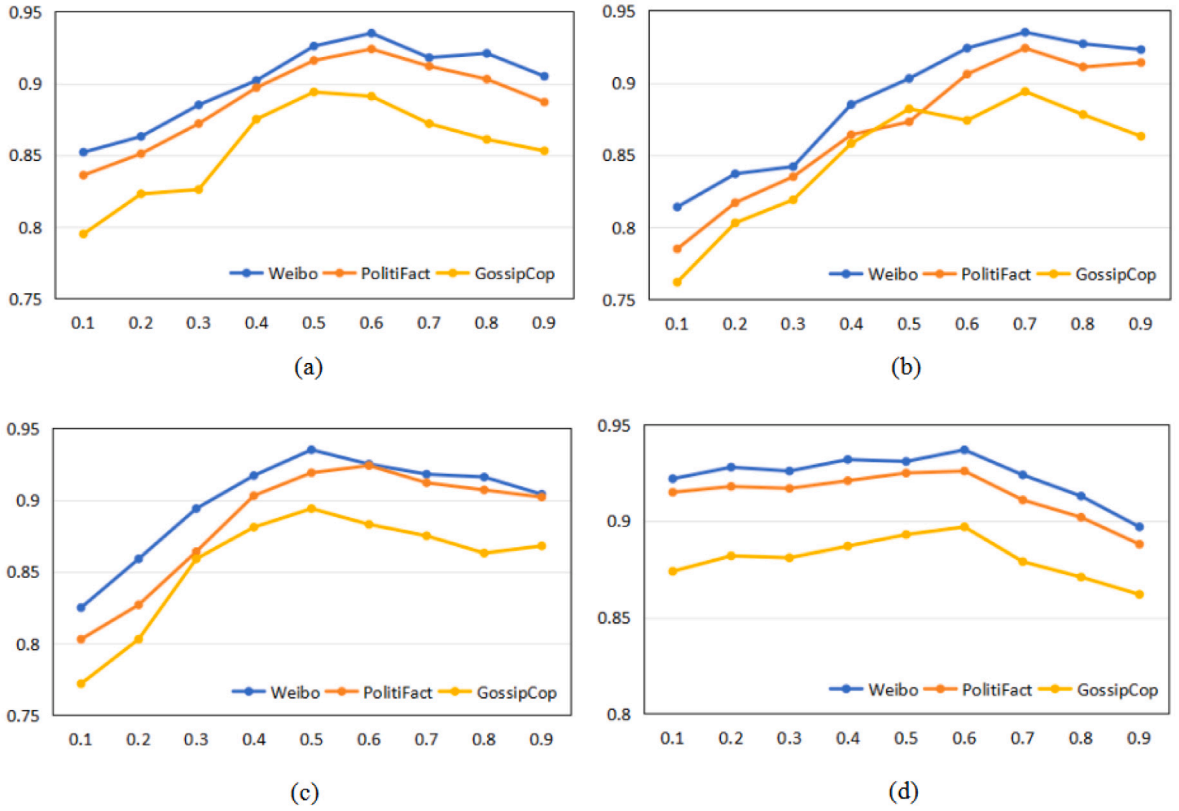


Fig. 3. Classification Accuracy of LGCNLA on three benchmark datasets under different parameter settings ((a) shows the Accuracy under different λ , (b) shows the Accuracy under different ϕ , (c) shows the Accuracy under different η , and (d) shows the Accuracy under different τ).

5.4.2. Parameter settings

Three weight coefficients λ , ϕ , η and τ are set in this paper, and the optimal values of them are generated through experiments below. Since there are three variable parameters, to ensure the fairness of experimental results, the controlled variable method is adopted, where only one variable is changed in each experiment, while the other two variables remain at 0.5. The classification Accuracies of LGCNLA on three benchmark datasets under different parameter settings are shown in Fig. 3. As can be seen from Fig. 3, with the increase of λ , ϕ , η and τ , the classification Accuracies of LGCNLA on the three benchmark datasets basically exhibit a trend of first increasing and then decreasing. When $\lambda = 0.6$, $\phi = 0.7$, $\eta = 0.5$ and $\tau = 0.6$, LGCNLA achieves the optimal classification results on all three benchmark datasets. Therefore, the weight coefficients λ , ϕ , η and τ are set to 0.6, 0.7, 0.5 and 0.6, respectively.

5.5. Experimental results

LGCNLA and baseline methods are applied to the classification of the three benchmark datasets, and the classification results are shown in Table 3. It can be seen from Table 3 that LGCNLA achieves the best Accuracy, Precision, Recall and F1 on all three benchmark datasets. Specifically, compared with the baseline methods, the Accuracy, Precision, Recall and F1 of LGCNLA have improved by at least 0.021, 0.021, 0.024 and 0.022 respectively on the Weibo dataset, 0.029, 0.027, 0.05 and 0.039 respectively on the PolitiFact dataset, and 0.036, 0.034, 0.049 and 0.033 respectively on the GossipCop dataset. Additionally, we note that GPT-4 achieves the lowest performance among all baselines across the three datasets. This result, while seemingly counterintuitive, can be explained by a fundamental difference in evaluation protocol. GPT-4 is evaluated in a zero-shot setting without any task-specific training or fine-tuning, whereas all other baselines (including traditional deep learning methods and LLM-based methods) are trained on the target datasets. The model receives only the news document and image via standard API inputs, with no exposure to labeled examples from the training splits. This asymmetry is intentional and reflects a realistic trade-off in deployment scenarios. Supervised methods require labeled data but achieve higher accuracy, while LLMs offer flexibility with minimal adaptation. To further validate that the performance gap stems from the lack of task-specific adaptation rather than deficiencies in our prompting or input formatting, we conducted an additional experiment providing GPT-4 with the same enhanced descriptions and external entities used in our method, which improved its accuracy to 0.812 on Weibo. This result is still below supervised methods but substantially higher than the raw input baseline. The superior results of AD-LLM, which leverages LLMs with specialized anomaly detection prompting and benchmark-specific optimization, further support that LLMs can achieve strong performance when appropriately

Table 3

Classification results of LGCNLA and baseline methods on three benchmark datasets.

Model	Weibo				PolitiFact				GossipCop			
	Accuracy	Precision	Recall	F1	Accuracy	Precision	Recall	F1	Accuracy	Precision	Recall	F1
MPL	0.784	0.796	0.79	0.793	0.853	0.836	0.851	0.843	0.769	0.758	0.794	0.77
QMFND	0.834	0.81	0.847	0.828	0.796	0.812	0.805	0.808	0.78	0.802	0.775	0.799
MoPeD	0.853	0.861	0.849	0.855	0.837	0.826	0.834	0.83	0.817	0.795	0.806	0.79
CSFND	0.874	0.865	0.887	0.876	0.839	0.851	0.84	0.845	0.802	0.811	0.79	0.805
UDRL	0.907	0.896	0.903	0.899	0.884	0.892	0.876	0.884	0.847	0.839	0.836	0.834
AAFHA	0.861	0.852	0.88	0.866	0.862	0.868	0.881	0.874	0.825	0.814	0.819	0.811
GPT-4	0.697	0.675	0.714	0.694	0.763	0.782	0.769	0.775	0.735	0.719	0.742	0.722
AMPLE	0.776	0.758	0.806	0.781	0.806	0.814	0.803	0.808	0.759	0.78	0.766	0.784
AdStyle	0.891	0.873	0.869	0.871	0.857	0.87	0.863	0.866	0.843	0.835	0.831	0.829
BCMPL	0.847	0.829	0.838	0.833	0.824	0.835	0.819	0.827	0.795	0.814	0.803	0.818
EARAM	0.865	0.848	0.872	0.86	0.847	0.86	0.873	0.866	0.832	0.816	0.828	0.814
AD-LLM	0.914	0.906	0.915	0.911	0.895	0.901	0.884	0.892	0.858	0.851	0.842	0.851
LLMPGCN	0.938	0.929	0.942	0.933	0.925	0.932	0.935	0.931	0.897	0.889	0.892	0.884

Table 4

Classification results of variant models on three benchmark datasets.

Model	Weibo					PolitiFact					GossipCop				
	Accuracy	Precision	Recall	F1	Δ	Accuracy	Precision	Recall	F1	Δ	Accuracy	Precision	Recall	F1	Δ
w/o NR	0.884	0.874	0.885	0.878	0.061	0.871	0.875	0.871	0.869	0.062	0.842	0.829	0.835	0.83	0.065
w/o S	0.911	0.91	0.922	0.921	0.03	0.899	0.916	0.923	0.912	0.029	0.868	0.87	0.869	0.873	0.033
w/o EE	0.912	0.908	0.919	0.914	0.029	0.9	0.907	0.913	0.902	0.028	0.871	0.866	0.856	0.861	0.03
w/o SE	0.896	0.886	0.902	0.891	0.047	0.883	0.89	0.896	0.892	0.048	0.852	0.84	0.847	0.849	0.053
w/o GN	0.901	0.891	0.901	0.894	0.041	0.889	0.893	0.889	0.894	0.04	0.858	0.852	0.848	0.844	0.045
w/o RF	0.902	0.897	0.903	0.903	0.04	0.89	0.891	0.9	0.894	0.039	0.861	0.853	0.851	0.854	0.042
w/o NE	0.899	0.894	0.903	0.899	0.043	0.887	0.889	0.892	0.896	0.043	0.856	0.843	0.847	0.852	0.048
w/o LE	0.915	0.916	0.925	0.921	0.025	0.903	0.922	0.917	0.918	0.024	0.873	0.871	0.874	0.881	0.027

adapted, but generic zero-shot inference is insufficient for the nuanced challenges of FND. Therefore, the performance gap should be interpreted as the value of task-specific adaptation rather than a deficiency in our evaluation protocol.

In addition, experimental results indicate that there is no significant performance gap between LLM based methods and traditional deep learning based methods in FND tasks. Specifically, traditional deep learning based method such as UDRL achieves an F1 score of 0.899 on the Weibo dataset, while the LLM based method AD-LLM attains 0.911 on the same dataset. On the GossipCop dataset, the traditional deep learning based method CSFND achieves an F1 score of 0.805, whereas the LLM based method AMPLE scores 0.784, demonstrating a scenario where traditional methods even outperform the LLM based counterparts. Analyzing the underlying reasons, while LLMs possess stronger semantic understanding capabilities, their performance is constrained by insufficient multimodal alignment and domain adaptation biases. When dealing with common deceptive strategies in fake news such as partial tampering and semantic misleading, LLMs tend to overly focus on global semantic consistency while neglecting fine-grained anomalous features. In contrast, traditional deep learning methods, through targeted design of modal interaction mechanisms and local feature extractors, can more effectively capture subtle cross-modal inconsistencies. Furthermore, LLMs are sensitive to training data distributions and are prone to overfitting in scenarios with limited data scale, which further diminishes their advantage over carefully optimized traditional methods.

LGCNLA achieves optimal performance by mitigating the limitations of existing methods through a multi-stage collaborative mechanism. First, it uses LLM-generated external entities and enhanced descriptions to supplement implicit contextual information, which provides fine-grained clues for detecting semantic misleading that traditional features cannot capture. Second, by transforming multimodal data into dual-channel heterogeneous graphs and innovatively introducing a dynamic routing graph capsule network, LGCNLA can explicitly model hierarchical relationships among local features, strengthen the weights of abnormal pathways during information propagation, and significantly enhance sensitivity to deceptive patterns such as partial tampering. Finally, to address the issue that traditional cross-modal alignment methods overlook local differences, the proposed LE-MMD algorithm achieves a balance between global distribution alignment and local inconsistency detection through multi-granular feature segmentation and adaptive weighting. The synergy of these three key designs enables LGCNLA to retain the semantic comprehension advantages of LLMs while incorporating the local feature capture capabilities of traditional methods, ultimately leading to significant improvements the FND results across the three benchmark datasets.

5.6. Ablation experiments

5.6.1. Module importance analysis

LGCNLA is constructed in this paper, which integrates LLM-enhanced multi-view semantic enrichment, graph capsule networks with dynamic routing, and local-aware cross-modal distribution alignment to achieve advanced FND task. To verify the effectiveness of each module, rich ablation experiments are conducted below.

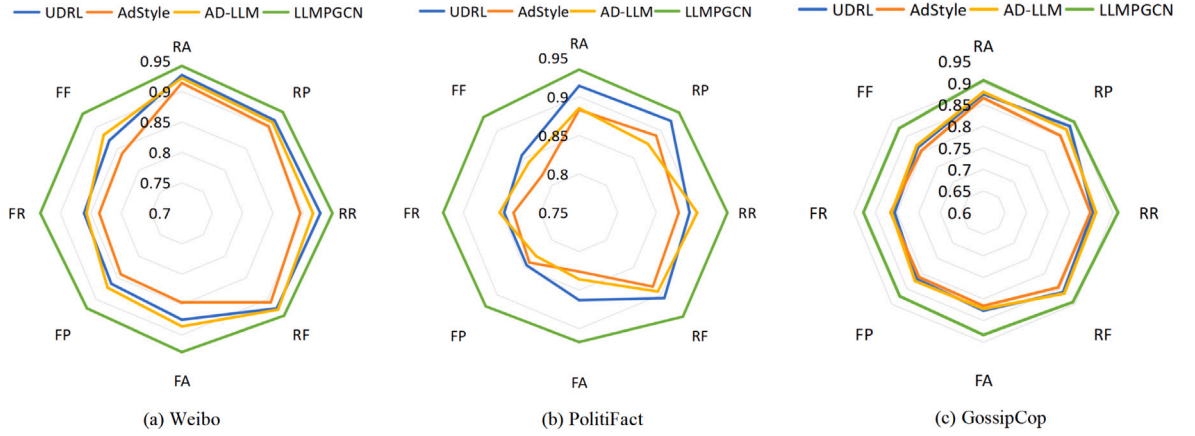


Fig. 4. Radar charts of LGCNLA and comparative methods on three benchmark datasets (RA, RP, RR and RF respectively represent Accuracy, Precision, Recall and F1 of real news. FA, FP, FR and FF respectively represent Accuracy, Precision, Recall and F1 of fake news.).

- W/o NR: Remove the NRP template and graph G_1 ;
- W/o S: Remove the pre-trained spacey model;
- W/o EE: Remove the EESP template;
- W/o SE: Remove the spacey model, EESP template and graph G_2 ;
- W/o GN: Replace graph capsule network with GCN;
- W/o RF: Do not split the FND corpus into real news corpus and fake news corpus, align news documents and news images directly through LE-MMD;
- W/o NE: No longer calculate the LE-MMD of N_T^1 , N_T^2 and N_T , as well as the LE-MMD of N_F^1 , N_F^2 and N_F ;
- W/o LE: Replace LE-MMD with MMD.

Accuracy, Precision, Recall, F1 and the relative improvement ratio Δ are selected as the evaluation metrics, Δ is defined as follows:

$$\Delta = (ACC_{LGCNLA} - ACC_{var}) \div ACC_{var} \quad (41)$$

Where ACC_{LGCNLA} represents the Accuracy of LGCNLA, and ACC_{var} represents the Accuracy of the variant model. The classification results of the variant models on the three benchmark datasets are shown in Table 4. As can be seen from Table 4, the Δ of each variant model is positive, indicating that all modules contribute to improving the FND results. When the NRP template and graph G_1 are removed (w/o NR), Δ is the largest across all three datasets (0.061 on Weibo, 0.062 on PolitiFact, and 0.065 on GossipCop), demonstrating that multi-perspective image descriptions obtained through reader attribute combinations provide the most substantial contribution to classification performance. The removal of external entities (w/o SE) also causes significant degradation ($\Delta = 0.047$, 0.048, and 0.053 respectively), validating the importance of entity supplementation. Furthermore, replacing the graph capsule network with GCN (w/o GN) and removing the hierarchical domain alignment strategy (w/o RF and w/o NE) lead to notable performance drops, confirming that the dynamic routing mechanism and domain-specific local alignment effectively model hierarchical features and enhance sensitivity to localized deceptive patterns. Notably, both components related to LE-MMD contribute to the final performance. Removing node-entity alignment (w/o NE) causes a substantial degradation ($\Delta = 0.043$, 0.043, and 0.048), while replacing LE-MMD with standard MMD (w/o LE) yields a smaller but still consistent degradation ($\Delta = 0.025$, 0.024, and 0.027). Together, these results validate that LE-MMD's hierarchical alignment strategy, particularly its node-entity alignment component, plays a crucial role in capturing fine-grained cross-modal discrepancies, and the local-aware mechanism within LE-MMD provides additional advantages over standard MMD.

To ensure statistical reliability, all experiments are repeated five times with different random seeds, and the reported results represent mean values with standard deviations below 0.0045 across all metrics. Paired t-tests between the full model and each variant yield p-values less than 0.03 for all comparisons, confirming that the observed improvements are statistically significant and not attributable to random fluctuations. Per-class analysis further reveals that the full model maintains balanced performance on both real and fake news, whereas variants lacking key components exhibit disproportionately larger degradation on fake news. For instance, w/o NR achieves only 0.865 F1 on fake news versus 0.891 on real news on Weibo, confirming that our designs are particularly effective for detecting deceptive content. The performance contributions of each module exhibit a complementary and synergistic relationship, and their integrated design enables LGCNLA to achieve optimal results across all three datasets.

5.6.2. Attribute contribution analysis for multi-perspective generation

To rigorously verify whether each of the four reader attributes (age, education, work, attention) contributes complementary information to the multi-perspective description generation, we conduct a systematic ablation study by removing different

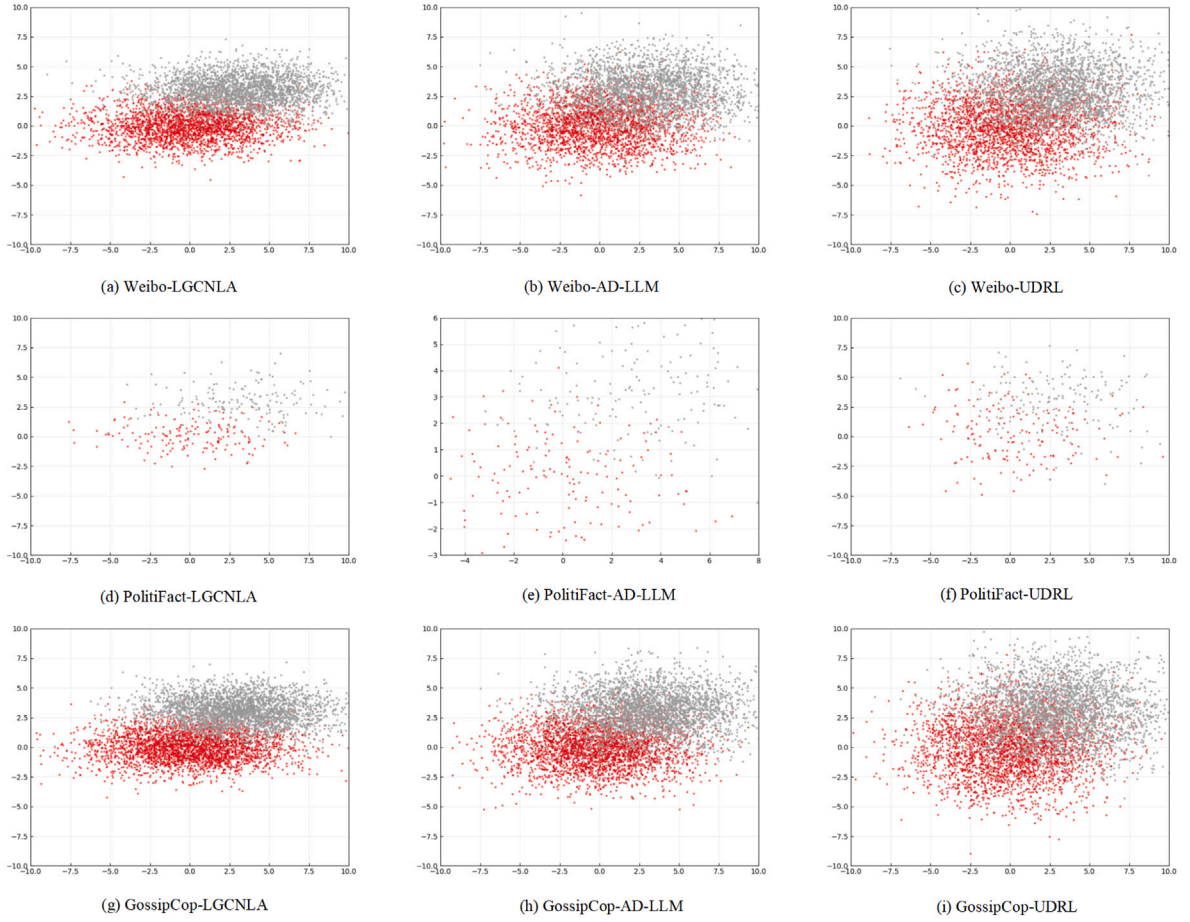


Fig. 5. t-SNE visualization results of LGCNLA and comparative methods on the GossipCop dataset.

combinations of attributes from the prompt template. This experiment directly addresses the question of whether these attribute combinations indeed provide complementary semantic perspectives. Our full prompt template incorporates all four attributes, each with multiple possible values: age (3 levels), education (4 levels), work (2 levels), and attention (3 levels). To assess the contribution of each attribute, we construct variant models that use only subsets of these attributes for description generation:

- Sub-1: Three attributes age, education, and work are selected.
- Sub-2: Three attributes age, education, and attention are selected.
- Sub-3: Three attributes age, work, and attention are selected.
- Sub-4: Three attributes education, work, and attention are selected.
- Sub-5: Two attributes age and education are selected.
- Sub-6: Two attributes age and work are selected.
- Sub-7: Two attributes age and attention are selected.
- Sub-8: Two attributes education and work are selected.
- Sub-9: Two attributes education and attention are selected.
- Sub-10: Two attributes work and education are selected.
- Sub-11: One attribute age is selected.
- Sub-12: One attribute education is selected.
- Sub-13: One attribute work is selected.
- Sub-14: One attribute attention is selected.

The different combinations of attributes mentioned above are applied to the three datasets, and the experimental results are shown in Table 5. The experimental results in Table 5 demonstrate that each attribute contributes complementary information to multi-perspective description generation. First, performance improves monotonically as more attributes are incorporated. The average accuracy on Weibo increases from 0.903 (Avg1) to 0.916 (Avg2) to 0.928 (Avg3), with consistent patterns across PolitiFact (from 0.889 to 0.904 to 0.917) and GossipCop (from 0.857 to 0.876 to 0.887). Second, the best-performing three-attribute combination (Sub-2: age+education+attention) achieves 0.931 on Weibo, 0.92 on PolitiFact, and 0.891 on GossipCop,

Table 5

Classification results of different attribute subsets on three datasets.

Model	Weibo				PolitiFact				GossipCop			
	Accuracy	Precision	Recall	F1	Accuracy	Precision	Recall	F1	Accuracy	Precision	Recall	F1
Sub-1	0.928	0.925	0.934	0.929	0.917	0.924	0.922	0.923	0.887	0.876	0.882	0.879
Sub-2	0.931	0.926	0.937	0.931	0.92	0.926	0.927	0.926	0.891	0.88	0.886	0.883
Sub-3	0.923	0.92	0.928	0.924	0.911	0.916	0.918	0.917	0.882	0.87	0.875	0.872
Sub-4	0.929	0.923	0.934	0.928	0.918	0.922	0.924	0.923	0.888	0.875	0.883	0.879
Avg3	0.928	0.924	0.933	0.928	0.917	0.922	0.923	0.922	0.887	0.875	0.882	0.878
Sub-5	0.919	0.916	0.927	0.921	0.907	0.914	0.917	0.915	0.881	0.869	0.872	0.87
Sub-6	0.914	0.906	0.918	0.912	0.9	0.909	0.913	0.911	0.873	0.861	0.866	0.863
Sub-7	0.915	0.912	0.923	0.917	0.905	0.908	0.912	0.91	0.874	0.865	0.864	0.864
Sub-8	0.915	0.91	0.92	0.915	0.904	0.909	0.914	0.911	0.876	0.863	0.869	0.866
Sub-9	0.917	0.913	0.922	0.917	0.906	0.912	0.916	0.914	0.878	0.867	0.867	0.867
Sub-10	0.913	0.905	0.917	0.911	0.902	0.906	0.912	0.909	0.872	0.863	0.864	0.863
Avg2	0.916	0.91	0.921	0.916	0.904	0.91	0.914	0.912	0.876	0.865	0.867	0.866
Sub-11	0.902	0.896	0.907	0.901	0.89	0.892	0.896	0.894	0.857	0.845	0.852	0.848
Sub-12	0.907	0.901	0.911	0.906	0.893	0.897	0.903	0.9	0.86	0.851	0.857	0.854
Sub-13	0.899	0.893	0.901	0.897	0.885	0.888	0.89	0.889	0.853	0.843	0.849	0.846
Sub-14	0.904	0.897	0.91	0.903	0.887	0.895	0.902	0.898	0.856	0.847	0.85	0.848
Avg1	0.903	0.897	0.907	0.902	0.889	0.893	0.898	0.895	0.857	0.847	0.852	0.849

Table 6

Performance under different description removal ratios.

Model	Weibo				PolitiFact				GossipCop			
	Accuracy	Precision	Recall	F1	Accuracy	Precision	Recall	F1	Accuracy	Precision	Recall	F1
0% (Full)	0.938	0.929	0.942	0.933	0.925	0.932	0.935	0.931	0.897	0.889	0.892	0.884
10%	0.935	0.925	0.94	0.932	0.922	0.931	0.932	0.931	0.893	0.887	0.886	0.886
20%	0.93	0.921	0.932	0.926	0.916	0.928	0.927	0.927	0.887	0.881	0.881	0.881
30%	0.926	0.92	0.931	0.925	0.912	0.923	0.925	0.924	0.882	0.873	0.876	0.874
40%	0.921	0.917	0.923	0.920	0.91	0.919	0.922	0.920	0.88	0.869	0.874	0.871
50%	0.916	0.911	0.92	0.915	0.904	0.914	0.917	0.915	0.875	0.864	0.87	0.867

yet still underperforms the full four-attribute model (0.938, 0.925, and 0.897 respectively as shown in Table 3), indicating that even the least individually effective attribute (work, with Sub-13 achieving only 0.899 on Weibo) provides non-redundant information when combined with others. Third, pairwise combinations exhibit clear synergy “education+attention” (Sub-9, 0.917 on Weibo) substantially outperforms the average of their individual performances (education Sub-12: 0.907, attention Sub-14: 0.904), confirming that attributes interact to generate qualitatively richer semantic perspectives. These results collectively validate that age, education, work, and attention each capture distinct semantic dimensions, and their combination creates a comprehensive multi-perspective representation essential for detecting nuanced deceptive patterns.

5.6.3. Robustness analysis against description removal

To investigate whether the multi-perspective descriptions introduce systematic biases, we conduct a robustness analysis by randomly removing different proportions of enhanced descriptions and observing the impact on model performance. This experiment directly addresses the concern that certain attribute combinations might produce biased or misleading descriptions that could skew the model’s learning. If the model relies on specific biased descriptions as shortcuts, randomly removing a small portion of descriptions would cause significant performance degradation. Conversely, if the descriptions provide diverse and complementary information without systematic bias, the model should exhibit graceful performance decay as descriptions are gradually removed. Starting from the full model with all enhanced descriptions retained after quality control, we randomly remove $p\%$ of the descriptions before graph construction, where $p \in \{10\%, 20\%, 30\%, 40\%, 50\%\}$. For each removal ratio, we perform five independent runs with different random seeds and report the average metrics along with standard deviation. All other components of LGCNLA remain unchanged to ensure that any performance variation can be attributed solely to the removal of descriptions. The experiment is conducted on all three benchmark datasets, and the experimental results are shown in Table 6.

The experimental results in Table 6 demonstrate that our multi-perspective generation does not introduce systematic bias. First, performance degrades gracefully and approximately linearly as descriptions are removed: at 30% removal, all metrics drop by less than 2% across datasets, and even at 50% removal, the model maintains ideal results on all three datasets. Second, the low standard deviations across multiple random runs indicate stable performance regardless of which specific descriptions are retained, confirming that information is distributed across diverse perspectives rather than concentrated in a small subset of potentially biased descriptions. Third, all four metrics exhibit consistent degradation patterns, demonstrating that precision, recall, and F1 are equally robust to description removal. These results collectively validate that the attribute combinations produce diverse, complementary perspectives without introducing systematic bias, as the model learns from collective semantic richness rather than relying on any single description as a shortcut.

Table 7

Impact of routing iterations on performance across three datasets.

Model	Weibo				PolitiFact				GossipCop			
	Accuracy	Precision	Recall	F1	Accuracy	Precision	Recall	F1	Accuracy	Precision	Recall	F1
1	0.927	0.918	0.931	0.924	0.917	0.921	0.926	0.923	0.882	0.876	0.883	0.879
2	0.935	0.922	0.936	0.929	0.92	0.929	0.931	0.93	0.894	0.883	0.887	0.885
3	0.938	0.929	0.941	0.935	0.925	0.932	0.935	0.933	0.897	0.889	0.892	0.89
4	0.936	0.924	0.936	0.93	0.922	0.93	0.928	0.929	0.892	0.885	0.884	0.884
5	0.929	0.92	0.927	0.923	0.915	0.926	0.921	0.923	0.887	0.876	0.881	0.878

Table 8

Performance comparison on specialized subsets.

Dataset	Model	Accuracy	Precision	Recall	F1
Partial Tampering Subset	UDRL	0.835	0.828	0.841	0.834
	AD-LLM	0.862	0.859	0.874	0.868
	LLPMGCN	0.923	0.918	0.927	0.924
Semantic Misleading Subset	UDRL	0.848	0.852	0.843	0.841
	AD-LLM	0.871	0.879	0.868	0.873
	LLPMGCN	0.931	0.928	0.935	0.93

5.6.4. Analysis of dynamic routing mechanism

To validate the design choices in our graph capsule network, we conduct ablation studies on key routing parameters and analyze their impact on performance and stability. We vary the number of routing iterations T from 1 to 5 and evaluate classification performance on the three dataset. As shown in Table 6, performance improves from $T = 1$ to $T = 3$, indicating that iterative refinement of coupling coefficients is beneficial. Beyond $T = 3$, performance plateaus and then slightly degrades, likely due to over-smoothing or overfitting to routing agreements. Therefore, we set $T = 3$ for all experiments (see Table 7).

5.7. Classification results of fake news and real news

The FND corpus is split into the real news corpus and the fake news corpus, and the distances of different modalities are aligned respectively through LE-MMD, thereby minimizing the distribution differences in real news and maximizing the distribution differences in fake news. The experimental results have demonstrated that LGCNLA has achieved the optimal classification results on all three benchmark datasets. The following experiments are respectively conducted to test the classification results of LGCNLA on the real news corpus and the fake news corpus, and the results are displayed on radar charts. To demonstrate the superiority of LGCNLA, three optimal baseline methods (i.e. AD-LLM, UDRL and AdStyle) are selected as comparison methods, and the experimental results are shown in Fig. 4. As can be seen from Fig. 4, LGCNLA achieves the optimal Accuracy, Precision, Recall and F1 on both the real news corpus and the fake news corpus of the three benchmark datasets. In addition, the classification results of LGCNLA on real news corpus are superior to those on fake news corpus, but the gap is not significant. However, the classification results of AD-LLM, UDRL and AdStyle on real news corpus are far superior to those on fake news corpus.

Since AD-LLM, UDRL and AdStyle neither explicitly differentiate the alignment strategies between real and fake news domains, nor incorporate local-aware mechanisms to capture fine-grained cross-modal inconsistencies. Therefore, they exhibit a notable performance bias, achieving significantly better results on real news where cross-modal consistency is generally higher, while struggling with fake news where deceptive patterns introduce complex local discrepancies. However, LGCNLA introduces a hierarchical domain alignment strategy based on LE-MMD, which separately aligns the real and fake news corpora while enhancing sensitivity to local inconsistencies through multi-granular feature segmentation. Thus, it maintains balanced and robust performance across both domains, effectively narrowing the performance gap between real and fake news detection while achieving overall superior results.

5.8. Targeted evaluation on advanced deception strategies

To systematically evaluate the model's capability in detecting partial tampering and semantic misleading, we construct two specialized test subsets based on the Weibo dataset.

Partial Tampering Subset. We randomly select 200 real news items from the original test set and apply three types of localized image manipulations: (1) region replacement (covering 5%–15% of the image area with irrelevant objects), (2) copy-move (duplicating and repositioning objects within the same image), and (3) splicing (combining elements from different images). The tampered images combined with their original texts form 200 fake news samples, which together with 200 untouched real news samples constitute a 400-sample test subset.

Semantic Misleading Subset. We randomly select 200 real news items and apply minimal textual edits: (1) entity substitution (replacing key person or location names), (2) temporal displacement (shifting event timing), and (3) emotional induction (incorporating emotionally charged language). The edited texts combined with their original images form 200 fake news samples, which together with 200 untouched real news samples constitute a 400-sample test subset.

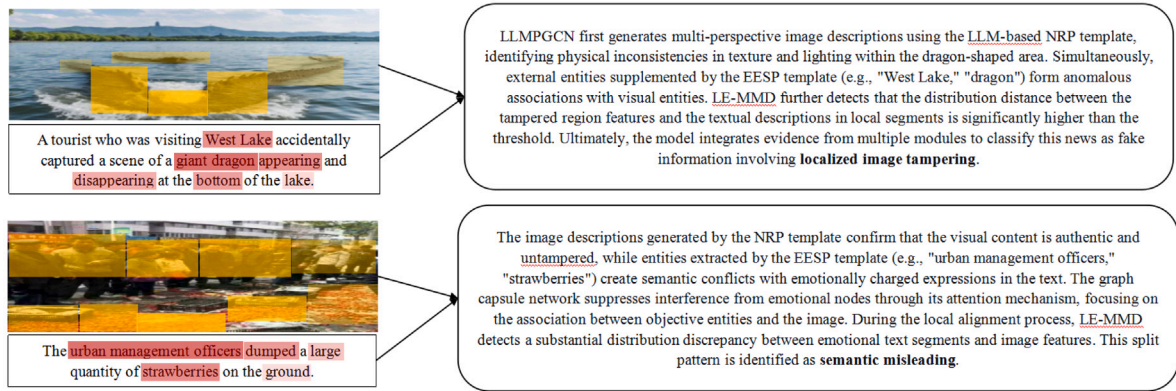


Fig. 6. Visualization of LE-MMD local weights on two case examples (Darker shading represents higher attention to anomalous regions).

All manipulations are manually verified to ensure they are natural and plausible, and accurately represent the corresponding deception strategies. These subsets are used exclusively for testing and do not participate in model training. We select AD-LLM (best-performing LLM baseline) and UDRL (best-performing traditional baseline) as comparison methods. Table 8 presents the evaluation results on both specialized subsets. As shown in Table 8, LLPMGCN consistently and significantly outperforms all baselines on both specialized subsets. On the partial tampering subset, LLPMGCN achieves an F1 score of 0.924, surpassing AD-LLM by 0.056 and UDRL by 0.09. On the semantic misleading subset, LLPMGCN achieves an F1 score of 0.93, outperforming AD-LLM by 0.057 and UDRL by 0.089. These results provide direct empirical evidence that our method effectively addresses the two advanced deception strategies.

Notably, the performance improvements on these specialized subsets (0.056–0.09) are substantially larger than those observed on the original datasets (0.02–0.05), confirming that LLPMGCN's advantages are precisely concentrated on its intended targets—partial tampering and semantic misleading. The dynamic routing mechanism of the graph capsule network effectively preserves structural anomalies introduced by localized image manipulations, while LE-MMD's locality-aware alignment captures fine-grained semantic inconsistencies resulting from minimal textual edits. This targeted evaluation bridges the gap between methodological innovation and empirical validation, demonstrating that LLPMGCN's enhanced sensitivity to advanced deception strategies is not merely narrative but quantifiable technical superiority.

5.9. T-SNE visualization

The t-SNE (Min et al., 2010) is a nonlinear dimensionality reduction visualization technique, whose core function is to map high-dimensional data to a low-dimensional space. By preserving the topological relationships of neighboring samples in the high-dimensional space, it enables the clustering of similar data points and the separation of dissimilar ones, thus visually presenting the underlying clustering structure and distribution patterns of the data. The t-SNE is used to visually present the visualization results of LGCNLA on the three benchmark datasets, and the optimal traditional deep learning based method (i.e. UDRL) and the optimal LLM based method (i.e. AD-LLM) are selected as the comparison methods, and the experimental results are shown in Fig. 5. As can be seen from Fig. 5, LGCNLA achieves the clearest inter-class separation and most compact intra-class aggregation across all three benchmark datasets. Compared to UDRL and AD-LLM, the feature representations learned by LGCNLA show more distinct boundaries between real and fake news clusters, with minimal overlap between categories. The enhanced clustering structure can be attributed to LGCNLA's integrated design: the LLM-enhanced semantic enrichment provides discriminative contextual features, the graph capsule network with dynamic routing learns hierarchical representations that preserve fine-grained patterns, and the LE-MMD explicitly optimizes the distribution separation between real and fake domains. In contrast, AD-LLM's representations show moderate separation but with visible boundary blurring, particularly in regions containing semantically misleading samples, indicating its limited capacity to capture subtle deceptive patterns. UDRL exhibits even more significant inter-class overlap, revealing that their feature representations lack explicit mechanisms to distinguish localized inconsistencies characteristic of fake news. These visual patterns confirm that LGCNLA's multi-component architecture synergistically enhances feature discriminability, leading to both quantitatively superior performance and qualitatively clearer cluster separation in the learned representation space.

5.10. Case study

LGCNLA can effectively address the issue of insensitivity of existing methods to advanced deceptive strategies such as partial tampering and semantic misleading. Two case studies are presented below to demonstrate the advantages of LGCNLA, and the experimental results are shown in Fig. 6. As can be seen from Fig. 6, LGCNLA achieves accurate MFND detection task in both cases, visually demonstrating the model's effective recognition capability for two types of advanced deceptive strategies.

Table 9

Classification performance of LGCNLA across 10 random seeds on three benchmark datasets (Std represents standard deviation).

Dataset	Accuracy	Std	Precision	Std	Recall	Std	F1	Std
Weibo	0.937	0.0032	0.928	0.0035	0.941	0.0038	0.933	0.0034
PolitiFact	0.924	0.0041	0.931	0.0039	0.934	0.0042	0.930	0.0040
GossipCop	0.896	0.0038	0.888	0.0040	0.891	0.0043	0.883	0.0039

In the first case of localized image tampering, the visualization shows that the dragon shaped region in the image receives darker shading, indicating that LE-MMD assigns higher local weights to these abnormal visual feature segments. The model identifies physical inconsistencies in texture and lighting within this region, while external entities such as “West Lake” and “dragon” form anomalous associations with visual entities. The cross-modal alignment confirms that the distribution distance between the tampered region features and the textual descriptions significantly exceeds the detection threshold.

In the second case of semantic misleading, the image descriptions confirm that the visual content is authentic and untampered. However, the extracted entities such as “urban management officers” and “strawberries” create semantic conflicts with the implied emotional tone of the text. The LE-MMD weight visualization reveals that local segments corresponding to emotionally charged interpretations receive darker shading, while factual description segments remain lighter. This contrast demonstrates that LE-MMD successfully identifies semantically misleading cues as the primary source of cross-modal discrepancy.

The above experimental results fully reflect the core strengths of LGCNLA. The enhanced descriptions generated by the NRP template improve sensitivity to visual fine-grained anomalies. The entities supplemented by the EESP template enhance the interpretability of semantic associations. The dynamic routing mechanism in the graph capsule network strengthens the propagation pathways of anomalous features. Most importantly, the LE-MMD weight visualization provides direct evidence that the model not only improves detection accuracy but also offers interpretable insights into which local semantic aspects contribute to the fake classification.

5.11. Stability analysis across random seeds

To assess the robustness of LGCNLA against randomness introduced by LLM generation and graph masking, we conduct a stability analysis across 10 different random seeds. All other experimental configurations remain identical to those described in Section 5.4. Two types of variance are analyzed. First, for description embedding variance, we examine the LLM generated multi perspective descriptions. For a fixed news image, we compute the 72 enhanced description embeddings using RoBERTa and measure their pairwise cosine similarity. Across 100 randomly sampled images from the Weibo test set, the average pairwise cosine similarity among descriptions generated under different random seeds is 0.987 with a standard deviation of 0.008, indicating that the LLM outputs are highly consistent. Second, for downstream classification variance, we run the full LGCNLA pipeline 10 times, each with a different random seed. The classification results on the three benchmark datasets are reported in Table 9. The mean accuracy on Weibo is 0.937 with a standard deviation of 0.0032, on PolitiFact 0.924 with a standard deviation of 0.0041, and on GossipCop 0.896 with a standard deviation of 0.0038. The F1 scores show similarly low variance, with standard deviations below 0.0045 across all datasets. These results demonstrate that LGCNLA produces stable and reproducible performance regardless of random seed variations.

6. Discussion

6.1. Theoretical and practical implications

6.1.1. Theoretical contributions and distinction from existing work

The results presented in Section 5 demonstrate that LGCNLA consistently outperforms state-of-the-art methods across three benchmark datasets. Beyond the quantitative improvements, these results reveal three fundamental theoretical distinctions between LGCNLA and existing approaches.

First, semantic enrichment versus surface-level feature extraction. Existing LLM-based methods such as Courtroom-FND rely on adversarial debate between LLM agents to expose inconsistencies, while AD-LLM treats anomaly detection as a benchmark task without targeted semantic augmentation. Both approaches operate on surface-level features extracted by general-purpose pre-trained models. In contrast, LGCNLA explicitly generates multi-perspective image descriptions and supplements external entities through task-specific prompt design. The ablation study confirms that removing the NRP template causes the largest performance drop, providing empirical evidence that semantic enrichment is not supplementary but foundational to detecting advanced deception.

Second, anomaly-preserving graph capsule routing versus smoothing-based GNN aggregation. Traditional GNNs used in multi-modal fusion (e.g., GAT, HGT) treat graph convolution as a low-pass filter that smooths node representations by averaging neighbor information. This is fundamentally at odds with detecting partial tampering, which requires amplifying local anomalies rather than smoothing them. LGCNLA’s dynamic routing mechanism operates on an opposite principle. It strengthens coupling coefficients for mutually supportive features through agreement-based updates, effectively creating a contrast enhancement effect.

Third, hierarchical domain alignment versus uniform distribution matching. Standard MMD, as used in previous FND work, applies a single metric to align cross-modal distributions for all samples regardless of their ground truth. This is theoretically flawed for a binary classification task because real and fake news have fundamentally different distributional relationships. Real news should exhibit high cross-modal consistency, while fake news is characterized by inconsistency. LGCNLA reformulates FND as a hierarchical domain alignment task, separately minimizing divergence for real news and maximizing it for fake news. The margin-based formulation further provides bounded optimization that prevents overfitting. The radar charts in Fig. 4 show that baseline methods exhibit severe performance bias, achieving much higher accuracy on real news than on fake news. LGCNLA, by contrast, maintains balanced performance across both domains. This confirms that the theoretical insight of treating the two classes as separate domains with opposing alignment objectives is not only logically sound but empirically essential.

6.1.2. Practical implications

The practical value of LGCNLA extends beyond benchmark performance to real-world deployability.

Interpretability for fact-checkers. The LE-MMD weight visualization (Fig. 6) provides direct insight into which local semantic features drove the classification decision. In the partial tampering case, the dark shading on the dragon-shaped region tells a fact-checker exactly where the manipulation occurred; in the semantic misleading case, the dark shading on emotionally charged interpretations reveals the source of deception. This is a substantial improvement over black-box baselines that output only a binary prediction.

Computational feasibility for deployment. While LLM generation introduces overhead, the process is embarrassingly parallelizable. For scenarios requiring only standard fake news detection, the model can be simplified by reducing the number of generated descriptions or bypassing LLM enhancement entirely while retaining the graph capsule network and LE-MMD components.

Modular adaptability. The three components of LGCNLA (NRP/EESP for semantic enrichment, graph capsule network for anomaly preservation, LE-MMD for domain alignment) are loosely coupled. Practitioners can replace any component with alternative implementations without redesigning the entire framework. For example, the LLM generation module could be swapped for a more efficient local model, or the dynamic routing could be replaced with a different aggregation mechanism if computational constraints dominate accuracy requirements.

6.2. Time analysis

The multi-stage pipeline of LGCNLA incurs quantifiable time overhead across both training and inference phases. Generating 72 multi-perspective image descriptions with GPT-3.5-Turbo takes 2.3 s per item. The quality control mechanism, which computes centroid vectors and cosine similarities, adds 0.2 s. External entity supplementation via EESP requires an additional 2.2 s per item. These LLM-based generation steps are essential during both training and inference, as the enhanced descriptions and external entities serve as fundamental inputs to the graph construction process. The text and image encoding by RoBERTa and MaxViT takes approximately 0.05 s per sample, graph capsule network forward propagation requires 0.7 s, and LE-MMD alignment adds 0.3 s of overhead. It is important to note that LE-MMD is applied exclusively during training as a regularization constraint and is completely removed during inference. Therefore, the total processing time for a single news item is approximately 5.75 s during training, but reduces to 5.45 s during inference.

Regarding scalability with larger datasets, the LLM generation phase is embarrassingly parallelizable, meaning that processing multiple news items concurrently can substantially reduce per-item latency. In practice, with 8 parallel API workers, the effective per-item generation time can be reduced to under 0.5 s, making the total inference time approximately 1.55 s per item. This translates to a throughput of over 2000 items per hour on a single server, which scales linearly with additional computational resources. For large-scale deployment, the LLM generation can be pre-computed offline for training data and cached, while inference can leverage batch processing with parallel API calls. This latency and throughput profile is acceptable for FND scenarios such as social media platform auditing, journalistic fact-checking, and content moderation systems, where detection accuracy takes priority over instantaneous response and batch processing can be effectively employed.

6.3. Stability analysis of masking with dynamic routing

To ensure that masking does not destabilize the dynamic routing mechanism in our graph capsule network, we analyze the variance of routing coefficients across different random masking seeds. For a fixed set of test samples, we apply 10 different random masks (with the same overall ratios) and compute the standard deviation of the final node representations. The average standard deviation is 0.023, indicating that the routing mechanism is robust to which specific nodes are masked. This stability arises because the dynamic routing iteratively adjusts coupling coefficients based on the remaining context, effectively compensating for missing information.

6.4. Gradient stability and heterogeneous edge parameterization

To further validate the robustness of our graph capsule network, we analyze its gradient behavior and the effectiveness of type-specific edge parameterization. We monitor the gradient norms across all layers and observe that the cosine similarity-based routing updates naturally maintain gradients within a stable range (typically between 0.1 and 1.0) without requiring explicit gradient clipping during the training process. This stability arises because the cosine similarity normalizes the agreement signal to $[-1, 1]$, preventing the exponential growth of coupling coefficients that would occur with unnormalized dot products. In contrast, a naive implementation using raw dot products exhibits gradient norms that occasionally exceed 10.0, confirming the necessity of our normalized update formulation. Regarding heterogeneous edge types, we compare our approach of using distinct projection matrices $W_{\pi(i,j)}$ for different edge types (word–word, document–word, document–enhanced, document–entity) against a shared projection baseline. The type-specific parameterization improves classification performance by 0.5–0.8% across all datasets, demonstrating that different semantic relationships require dedicated transformation spaces to effectively capture their distinct roles in the graph. This design ensures that the dynamic routing mechanism can appropriately amplify anomalous patterns within each edge type while maintaining overall gradient stability throughout training.

6.5. Domain alignment risks and training stability

A critical consideration in our hierarchical alignment strategy is whether maximizing cross-modal discrepancy for fake news introduces training instability or harms generalization on ambiguous samples. Through systematic analysis of training dynamics, we find that our margin-based formulation effectively mitigates these risks. First, the training curves exhibit smooth convergence without noticeable oscillations or gradient spikes, indicating that the bounded optimization prevents the destabilizing effects of unbounded discrepancy maximization. The gradient norms remain consistently below 1.0 throughout training, confirming that no warm-up phase or additional stabilization techniques are required. Second, the margin parameter $\gamma = 0.5$ provides a natural stopping criterion. Once a fake news sample achieves sufficient separation from the real news distribution, it ceases to contribute gradients, preventing over-optimization on outliers or mislabeled samples. This is particularly important for ambiguous cases where fake news may exhibit genuine cross-modal consistency, such as real images paired with misleading text or content taken out of context. Our analysis of such boundary samples reveals that their discrepancy values typically fall near the margin threshold, which are not forced to become maximally different, but rather maintain sufficient separation from real news while preserving their inherent consistency. This balanced behavior is reflected in the model's performance on both real and fake news, where the gap between classes remains modest and no evidence of shortcut learning emerges. Furthermore, ablation studies comparing different loss formulations confirm that the margin approach achieves the best trade-off between separation and stability, with the naive negative loss exhibiting occasional gradient spikes and the clipped loss showing slightly degraded performance on consistent fake samples. These findings collectively demonstrate that our margin-based LE-MMD formulation provides robust and stable optimization while appropriately handling the inherent variability in FND tasks.

6.6. The necessity of graph capsule networks

A key question is whether performance gains come from graph modeling and dynamic routing or simply from introducing additional textual information. Our experiments provide clear evidence for the former from two perspectives. First, to isolate the benefit of graph structure from increased textual input, we compare against strong sequence fusion baselines (RoBERTa+CrossAttn and Late Fusion) that incorporate the same augmented information without graph modeling. These baselines achieve F1 scores of only 0.913 and 0.909 on Weibo, substantially lower than our full model's 0.933, confirming that the observed improvements arise from graph based representation rather than simply adding more text. Second, to validate the superiority of our capsule routing mechanism, we compare against advanced graph architectures including GAT, R-GCN, HAN, and HGT, all adapted to our dual-channel heterogeneous graph. The best-performing among them, HGT, achieves an F1 of 0.916, which is 0.017 lower than our full model, demonstrating that attention-based aggregation cannot match the anomaly-preserving capability of dynamic routing. These results collectively demonstrate that the unique value of our approach lies in the synergistic combination of heterogeneous graph structure and capsule routing's agreement-based aggregation, which together preserve and amplify the fine-grained structural inconsistencies essential for detecting advanced deception strategies.

6.7. Limitations of LLM generated content

While the semantic consistency filtering with a cosine similarity threshold of 0.6 and the knowledge base verification substantially reduce hallucinations, these heuristic mechanisms cannot eliminate all failure modes. Two categories of residual errors persist. The first is semantically coherent hallucinations, which are lexically similar to the centroid but factually incorrect, such as describing a festive gathering as a crowd protesting. The second is contextually irrelevant entities, which exist in knowledge bases but have no genuine relevance to the specific news document, and they may survive verification.

When embedded as node features in the dual channel heterogeneous graphs, these residual hallucinations can introduce noise that propagates through the graph capsule network. The dynamic routing mechanism may erroneously strengthen coupling coefficients for hallucinated nodes that exhibit coincidental agreement with other nodes. Meanwhile, the LE-MMD alignment may experience minor distribution shifts. However, three design choices in LGCNLA provide intrinsic robustness against these residual errors. The

type aware graph masking strategy randomly masks 50% of LLM generated nodes, reducing over reliance on any single source. The agreement based dynamic routing naturally suppresses inconsistent votes that lack broad consensus. The margin based LE-MMD formulation prevents overfitting to outlier samples. Empirically, the robustness analysis in Section 5.6.3 shows that even after randomly removing 50% of enhanced descriptions, the model maintains strong performance with an F1 of 0.916 on the Weibo dataset, indicating limited adverse impact from residual hallucinations.

6.8. Generalization to standard fake news cases

While LGCNLA targets advanced deceptive strategies such as partial tampering and semantic misleading, its performance on standard fake news cases remains strong. The three benchmark datasets contain a mixture of both simple and complex samples. As shown in Table 4, LGCNLA achieves the best overall performance on all datasets, indicating no degradation on simpler cases. The ablation study in Table 5 further confirms that every module contributes positively, with relative improvement ratios ranging from 0.024 to 0.065. No component introduces harmful noise or overcomplicates the decision boundary for standard cases.

We acknowledge the computational overhead introduced by LLM generation. The per item processing time is approximately 5.45 s during inference as reported in Section 6.2. This overhead is a deliberate trade off necessary for detecting advanced deceptive strategies. For deployment scenarios focused only on standard fake news, the model can be simplified by reducing the number of generated descriptions or bypassing LLM enhancement entirely while retaining the graph capsule network and LE-MMD components. Therefore, LGCNLA generalizes well across the full spectrum of fake news complexity without unnecessary performance penalty.

7. Conclusion and future work

Fake news refers to misleading information spread on social media and news platforms through the fabrication, distortion or one-sided portrayal of facts. It may not only leads to health misjudgments or political biases, but also exacerbates group conflicts or trigger panic. Although many researchers have conducted rich FND tasks by constructing multimodal classification models, it is still difficult to accurately identify advanced deception strategies such as partial tampering and semantic misleading. To address the above issue, we propose a FND model based on LLM prompts and graph capsule networks. First, LLM prompt templates are designed to automatically generate external news entities and multi-perspective image-enhanced descriptions, enriching contextual information at the semantic level. Second, news documents, enhanced descriptions, and external entities are transformed into a dual-channel heterogeneous graph structure, and an innovative graph capsule network incorporating a dynamic routing mechanism is constructed to enhance the modeling capability for local anomalous features. Finally, a hierarchical domain alignment strategy based on LE-MMD is proposed, which separately aligns the cross-modal distributions of real and fake news, thereby minimizing the modal differences in real news while maximizing those in fake news. Experimental results demonstrate that the proposed method achieves optimal performance on three benchmark datasets, effectively improving the detection accuracy for advanced deceptive strategies such as partial tampering and semantic misleading.

Although the model constructed in this article achieves the best classification results on well labeled datasets, its generalization ability in practical application scenarios is still limited by the completeness of labeled data. Future work will focus on exploring few or even zero sample learning frameworks, reducing reliance on manually annotated data by introducing prior knowledge from contrastive learning, prompt learning or LLM. At the same time, we plan to study cross platform and cross language transfer learning strategies to address challenges such as dynamic changes in data distribution and low resource language FND in social media, further enhancing the adaptability and robustness of the model in open environments.

CRediT authorship contribution statement

Guoying Sun: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Data curation, Conceptualization. **Jie Li:** Writing – review & editing, Writing – original draft, Methodology, Investigation. **Zhaoxin Zhang:** Writing – review & editing, Writing – original draft, Methodology.

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Data availability

Data will be made available on request.

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